

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

STATISTICAL TECHNIQUES IN PREDICTING THE WINNERS OF PROFESSIONAL WOMEN TENNIS IN GRAND SLAMS BETWEEN THE YEARS 2001-2022

Master Thesis

by

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CHAPTER 1. INTRODUCTION

Statistical analysis may be the most attractive and intelligent way to ever extract and gain knowledge from data at hand. With statistical analysis, we can summarize data, find KPIs, observe patterns, and make predictions. This can be done in many fields, including healthcare, social media, and also sports. In this work, the focus is on sport analytics. Sport analysis is important because it gives insights of performance for teams and players to improve, allows for better understanding of the game, thereby, allowing for improving it and its rules, allows for more profitable betting, and provides satisfaction for fans to expect results and to have beliefs in the teams and players they support. Specifically, the work is done on data from the Women Tennis Association, WTA, tournaments because women's tennis is the fastest growing sport in the last years [1] and it generates very large revenues. This makes it an excellent choice for recommendations and improvements. Tennis, also, is a game of two players with the rules of the game commonly known. In other words, it is not a game that allows for too much strategies in-play but rather it favors the better performers. This makes prediction from its data less subjective and more explainable. Another reason, is that most of the available analysis is done on men's data and not on women's data.

Looking at previous works in analytics, Morgulev suggested an investigation of several methods of analysis that are driven by data for a wide variety of sports, both prior to and after the introduction of the era of computerization [2]. The popularity of using statistical methods to predict the winners of various tournaments is increasing [3] and the same is enabled by two factors at the least. The first factor lies in the fact that the results are apparent. This is in contrast with the predictions for the financial markets in which the variables of interest may lie hidden necessitating the use of volatility or proxy. Secondly, the large number of variables – like players' measurements and courts' surfaces – that could influence the result of any tournament broadens the range of statistical tools available to us. Variables of players as well as courts can have that effect.

Some factors of performance are cited in Kovacs work, where he explains the importance of training in a certain fashion to improve performance. According to Kovacs' work here, the performance is affected by speed and agility, stamina of the player, strength of the player, and the flexibility of the player [4]. In another of his work, Kovac explains the importance of deceleration in tennis. According to Kovac, dynamic balance, eccentric strength, power, and reactive strength majorly affect deceleration of the player; and the better a player is at deceleration the

better the performance due to better lateral movements, direction changes, and becoming less injury prone [5]. This next study, also from Kovac, focuses on firstly anthropometric data and advises to focus training on factors other than aerobic factors. This is because the study shows that tennis is not a purely aerobic sport and hence, the training should not be purely aerobic but rather have training breaks in between sets [6]. In general, the focus was on the difference between types of trainings, fluid injections, hormonal balances, and recovery factors. Other works focus on the players mental resilience by applying statistical methods after obtaining data from questionnaires [7], and not from in-game performances that can prove to be more predictive.

The main aim of this work is to analyze the data of WTA tournaments from 2001 till 2022 to predict the performances of players and the resulting winners and losers in Grand Slams; this is the general aim of the work. A competition is a Grand Slam when it is one of the following four competitions, namely: Australian Open, French Open, Wimbledon, and US Open. There are 3 sets per match and 7 rounds of single-elimination games in each Grand Slam competition (1st round, 2nd round, 3rd round, 4th round, quarter-final, semi-final, and final), totaling 127 matches (and 128 participants). Grand Slam tournaments now use a 32-seed structure to

seed players based on their rankings. Three objectives are defined to this end and interesting questions regarding player characteristics are raised and answered to support or reject commonly thought hypothesis.

The first hypothesis suggested, was that anthropometric data – height and age considered in this work – have an effect on performance. The hypothesis is that taller and younger players do better. The first objective, hence, focuses on that. For height, the first question to answer was its importance. After that, the focus was on seeing the different rankings and number of aces for different heighted players. Then, the analysis went on to see if there is an increase or decrease in win probability with every centimeter of increase in height and finally, the evolution of players' heights over time. For age and experience, the aim was to see the change in winning probabilities for different age groups, and for different age groups against each other. Then, the focus moved onto checking age advantages in last sets and finals for older versus younger players and the optimal years of experiences.

The second hypothesis was that players perform differently in different tournaments because different tournaments have different surfaces. So, the second objective, focused on studying serves and aces percentages on different surfaces. Specifically, the number of aces, the

percentage of first serves won, the percentage of serve points won, and the aces per serve. This would also show what kind of play works better on what kind of surfaces, and the percentage of wins for different clusters of players.

The third hypothesis was that there are concentration losses in final sets due to fatigue and overconfidence. Therefore, the third objective studied the sport psychology and how it affects players' performance. This included mental resilience of the players and how they perform under pressure. For this, there was an analysis of upsets between differently ranked players and the comebacks of players in different matches; this also checked if upsets happen more in Grand Slams or in smaller tournaments. Also, for upsets, the matches were analyzed to see how many sets (two or three) are more common, and what surfaces influence more upsets. For comebacks, first, the percentage of comebacks happening is calculated. Then, the focus was on what surfaces have more comebacks happening and what category of players perform more comebacks by checking the number of wins for players that make more comebacks.

In general summary, the objective of the analysis is to find insights from the data given in order to predict winning and losing of future WTA games beforehand. In this section, the objectives that were actually

analyzed were seen. These objectives focused on what a tennis fan would like to predict. That is to say, they were chosen in forms of questions a regular watcher would like to guess. Any tennis fan would like to know if their favorite player is improving over the years, doing well in a certain tournament, or is favorited to win a competition.

CHAPTER 2. METHODS

Starting with the design of this work, the approach taken is a quantitative approach that allows to understand and describe the variables from the dataset, before testing the suggested hypothesis, and coming up with models to explain. A cross-sectional study was conducted with proper data for WTA tennis. The work is descriptive and predictive; this is to describe characteristics, averages and trends, while allowing to test whether variables are related and whether results can be predicted. Next in the design is to identify the population and sampling method. The population is made up of all the tennis players playing in Grand Slams and the sample is made up of the tennis players playing in Grand Slams from the years 2001 until 2022. The sample method is a non-probability sample where we split our datasets into training and testing datasets to be able to test our models on the testing datasets.

For the data collection, this study looks at 22 years of Grand Slam singles matches for women (2001–2022) and use data from GitHub (49 variables) curated by Jeff Sackmann at **https://github.com/JeffSackmann** The dataset is rich with attributes about the players – winners and losers – in every round of every tournament played in the 22 years. In addition to

that more data was added using data scraping from www.tennisabstract.com/cgi-bin/leaders_wta.cgi.

2.1. DATASET

The variables from both sources are well explained under this section in summary tables containing the name of the variable, an explanation of it, and a specification of its type. Below is the first summary table.

Name of Variable	Description of Variable	Variable Type
tourney_id	a unique identifier for each tournament	Categorical
tourney_name	Name of tournament	Categorical
surface	Different surface type (clay, grass, etc.)	Categorical
draw_size	number of players in the draw	Numerical
tourney_level	Identifies the tournament type (G: grand slam)	Categorical

tourney_date	Date formatted as YYYYMMDD	Categorical
match_num	A match-specific identifier	Categorical
winner_id	the player_id used in this repo for the winner of the match	Categorical
winner_seed	Ranking of the player at the year of tournament	Numerical
winner_entry	Identifier for player entry ('WC' = wild card, 'Q' = qualifier, etc.)	Categorical
winner_name	Name of the winner	Categorical
winner_hand	Preferred hand of player (R = right, L = left, U = unknown)	Categorical
winner_ht	height in centimeters	Numerical

winner_ioc	three-character country code	Categorical
winner_age	age, in years, as of the tourney_date	Numerical
loser_id	the player_id used in this repo for the loser of the match	Categorical
loser_seed	Ranking of the player at the year of tournament	Numerical
loser_entry	Identifier for player entry ('WC' = wild card, 'Q' = qualifier, etc.)	Categorical
loser_name	Name of player that lost	Categorical
loser_hand	Preferred hand of player (R = right, L = left, U = unknown)	Categorical
loser_ht	height in centimeters	Numerical

loser_ioc	three-character country code	Categorical
loser_age	age, in years, as of the tourney_date	Numerical
score	Final score of sets of matches	Numerical
best_of	'3' or '5', indicating the the number of sets	Categorical
Round	Round in tournament (eg semi-final)	Categorical
minutes	match length	Numerical
w_ace	winner's number of aces	Numerical
w_df	winner's number of doubles faults	Numerical
w_svpt	winner's number of serve points	Numerical
w_1stIn	winner's number of first serves made	Numerical

w_1stWon	winner's number of first-serve points won	Numerical
w_2ndWon	winner's number of second-serve points won	Numerical
w_SvGms	winner's number of serve games	Numerical
w_bpSaved	winner's number of break points saved	Numerical
w_bpFaced	winner's number of break points faced	Numerical
l_ace	loser's number of aces	Numerical
l_df	loser's number of doubles faults	Numerical
l_svpt	number of serve points	Numerical
l_1stIn	number of first serves made	Numerical
l_1stWon	number of first-serve points won	Numerical

l_2ndWon	number of second-serve points won	Numerical
l_SvGms	number of serve games	Numerical
l_bpSaved	number of break points saved	Numerical
l_bpFaced	winner's number of break points faced	Numerical
winner_rank	winner's WTA rank as of the tourney_date	Numerical
winner_rank_points	number of ranking points	Numerical
loser_rank	Loser’s WTA rank as of the tourney_date	Numerical
loser_rank_points	number of ranking points	Numerical

Table 1 - First Variables Summary Table with Description and Type for Every Attribute

The next summary table if for the variables that were scraped.

Name of Variable	Description of Variable	Variable Type
Rk	Ranking	Numerical
Player	Name of player	Categorical
M	number of matches played	Numerical
M W-L	splits the number of matches in wins and losses	Numerical
M W%	percentage of wins	Numerical
SPW	serve points won	Numerical
SPW-InP	serve points won excluding aces and DFs	Numerical
Aces	Number of aces	Numerical
Ace%	Percentage of aces	Numerical
DFs	double faults	Numerical
DF%	percentage of double faults	Numerical

DF/2s	double fault per second serve	Numerical
1stIn	First serve percentage	Numerical
1st%	first serve percentage success	Numerical
2nd%	Second serve percentage success	Numerical
2%-InP	second serve success excluding DFs	Numerical
Hld%	hold percentage (% of service games won)	Numerical
Pts/SG	Points played per service game	Numerical
PtsL/SG	Points lost per service game	Numerical

Table 2 - Second Variables Summary Table with Description and Type for Every Attribute

The ranking appears to be a reliable measure for gauging players' strength, based on how Grand Slam events seed the bracket. This study takes this idea and turns it into a rating index for forecasting tennis match

outcomes. The Women's Tennis Association's current ranking algorithm is used to determine seeding of players and qualification for event admission. The stage of the competition they reached and the reputation of the tournament they competed in for the previous 52 weeks determine the players' ranking points. If a player has not played in the previous 52 weeks, their ranking points will be deducted. Tennis tournament ranking points can be obtained on the WTA's official websites. To understand how the data is correlated and how some hypothesis can be answered, exploratory data analysis is done in the next section.

2.2. STATISTICAL ANALYSIS

In this work, descriptive analysis which is mainly bivariate, is done on to understand how the performances are affected by certain attributes. This is tackled under bivariate analysis. Then some multivariate analysis is done and the methods are explained in a different section.

Additional statistical analysis for the second and third objectives is addressed here. For the second objective where the surfaces of different tournaments are studied, a Kruskal-Wallis test and its effect size are computed to see if there is any statistical difference between each metric used and the three different surfaces. For the third objective where the

effects of sports psychology are studied, a Kolmogorov-Smirnov test is done to check statistical differences in the two distributions of the plots for the advantages of age in the finals and ages respectively.

2.2.1. Bivariate analysis

To understand the data better, some exploratory data analysis is performed on the datasets. From the scrapped data, a plot to show the percentage of continuous and discrete columns is seen, as well as the complete and missing column percentages. Then, frequency and density plots are seen for the continuous variables. Moving to the dataset from Jeff Sackmann, there were the same plots for the continuous variables and added some for the discrete that gave some insight. Then, some categorical variables were plotted, specifically the surface variable, the tourney_name variable, and the preferred hands for both losers and winners.

To tackle the objectives, the bivariate methods used will be described here. For the first objective of anthropometric data versus wins, most of the analysis done was rather descriptive than modelling. This is because it is important to first see if the height and age indeed affect performances. For the height analysis, to see if changes in heights matter, a simple plot is made to see the distribution of the height difference

between two opponents. Once the idea of height advantage or disadvantage was obtained, some more plots were done to prove the idea further. There is a plot made for the number of aces against the different heights of players. The idea here is to see if taller players serve more aces, and in turn win more points and probably matches. The same thing was done to actually see the rankings of players with different heights. The idea here was to prove if players of a certain height group have lower (better) rankings than those in other height groups. For height, also, the probability of winning versus one-centimeter increases was seen and then finally, the evolution of players' average heights over the years.

More bivariate analysis is then performed on the age attributes to see how it can pose an advantage or a disadvantage. Age groups are defined and correlation plots for different age groups are seen. The idea is to identify the age groups that have the most winning advantages and then checking winning probabilities of when the age groups face each other with another correlation plot. After that, the idea of experience was added. The point of this was to see if older age gives an advantage due to experience and so an effort to see how older players do in finals and in last sets is made. For this, two distributions are obtained and then tested for statistical differences.

Realizing that experience does not always mean older players since some players start at older ages than others professionally, a density plot is made to see what is the optimal age of experience to reach the finals.

For the second objective, the match-based objective, the focus was on the different surfaces in different tournaments of the Grand Slams. Here, the first analyses that were performed were bivariate analysis – specifically a scatter plot – to compare the number of aces that are served on different surfaces and hence, come up with the surface that allows for faster serves. Then similar analysis was done for each of the following on different surfaces: the percentage of first serves won, the percentage of first serve points won, and finally the ratio of aces per serves.

For the third objective of finding if there is an effect on match results due to psychology factors, two things were looked at: upsets and comebacks. Starting with upsets, bivariate analysis was done first to understand if upsets happen more on certain surfaces, in Grand Slams or in smaller tournaments, and on which sets, mostly. After that, logistic regression modelling is performed to see how many sets it takes – two or three – to perform an upset, but this is tackled in the multivariate section after this. This aimed to prove if more comfortable players actually need only two sets and if players in need of performing an upset will do so in

one more set. For the second part of this objective, the comebacks, also bivariate analysis is performed to produce plots that prove on which surfaces comebacks happen more, and then with a scatter plot, which category of players perform more comebacks.

2.2.2. Multivariate Analysis

After the bivariate analysis was performed, some multivariate analysis is done. Starting with clustering, six clusters are done for three different surfaces. The results will show the percentage of wins for every cluster, the average height in every cluster, and finally, a plot of the Principal Component Analysis, PCA, of the clusters.

Next, to predict the winner and losers of the games from the data we have, some models are created. For a binary target variable called win/lose, the following columns are used as factors: "surface", "round", "tourney_level", "diff_seed", "age_diff", "ht_diff", "rank_diff", "experience", "last_result". The models created are logistic regression, random forest, decision tree, LDA, and Extreme Gradient Boosting. They are tested on a ‘test’ dataset that was not used to create the model and the statistical results and the confusion matrix results are seen and explained in the results section.

For logistic regression, the formulas are displayed below.

Suppose that we can find $\beta_0, \beta_1, \beta_2, \dots, \beta_p \in R^{p+1}$ such that for all observation $x \in R^p$,

$$\text{logit}\big(P(\text{win_lose} = 1|x)\big) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \quad (1)$$

where for all $\eta \in]0,1[$,

$$\text{logit}(\eta) = \log\left(\frac{\eta}{1-\eta}\right) \quad (2)$$

Thus, the probability η is give by:

$$P(\text{win_lose} = 1|x) = \frac{e^{\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_p x_p}}{1+e^{\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_p x_p}} \quad (3)$$

For decision trees, there isn't an explicit formula for its model but every split is based on the entropy which is amount of information needed to accurately describe the same sample. So, if a sample is homogeneous, it means all the element are similar, then entropy is 0, else if sample is equally divided than entropy is maximum 1.

$$\text{entropy} = -\sum_{i=1}^{n} p_i \log(p_i) \quad (4)$$

There is also the Gini Index/Gini impurity, which is the measure of inequality in sample. It has value between 0 (means the sample is perfectly homogeneous and all elements are similar) and 1 (means the maximal inequality among elements. It is the sum of the square of the probabilities of each class).

The Gini index is given by:

$$\text{Gini Index} = 1 - \sum_{i=1}^{n} p_i^2 \quad (5)$$

For Linear Discriminant Analysis – which aims to maximize the separation between two or more groups – the formula for combining two variables X_1 and X_2

$$\text{LD} = \alpha_1 X_1 + \alpha_2 X_2 \quad (6)$$

Where α_1 and α_2 are optimal values of the weight.

CHAPTER 3. RESULTS

3.1. EXPLORATORY DATA ANALYSIS (EDA)

For this work, a lot of hypotheses from the objectives were answered with descriptive and exploratory analysis. These will be displayed and explained in this section. The other methods are answered in the last section of the results and explained there.

Starting with exploratory data analysis about the data we have, in general, we start off with the parameters from the scraped data.

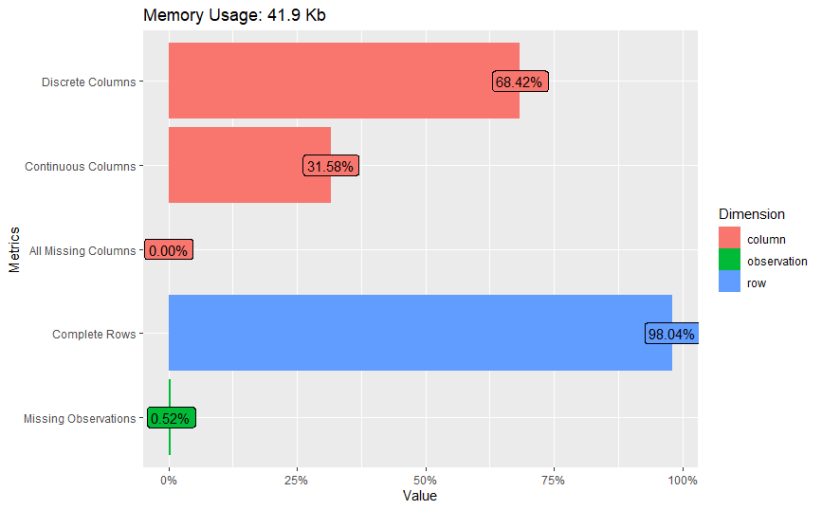


Figure 1 – Metrics of Variables (Scraped Data)

From the above we know that most of the variables are discrete ones and we have mostly complete data. Next, we take a look at some continuous variables.

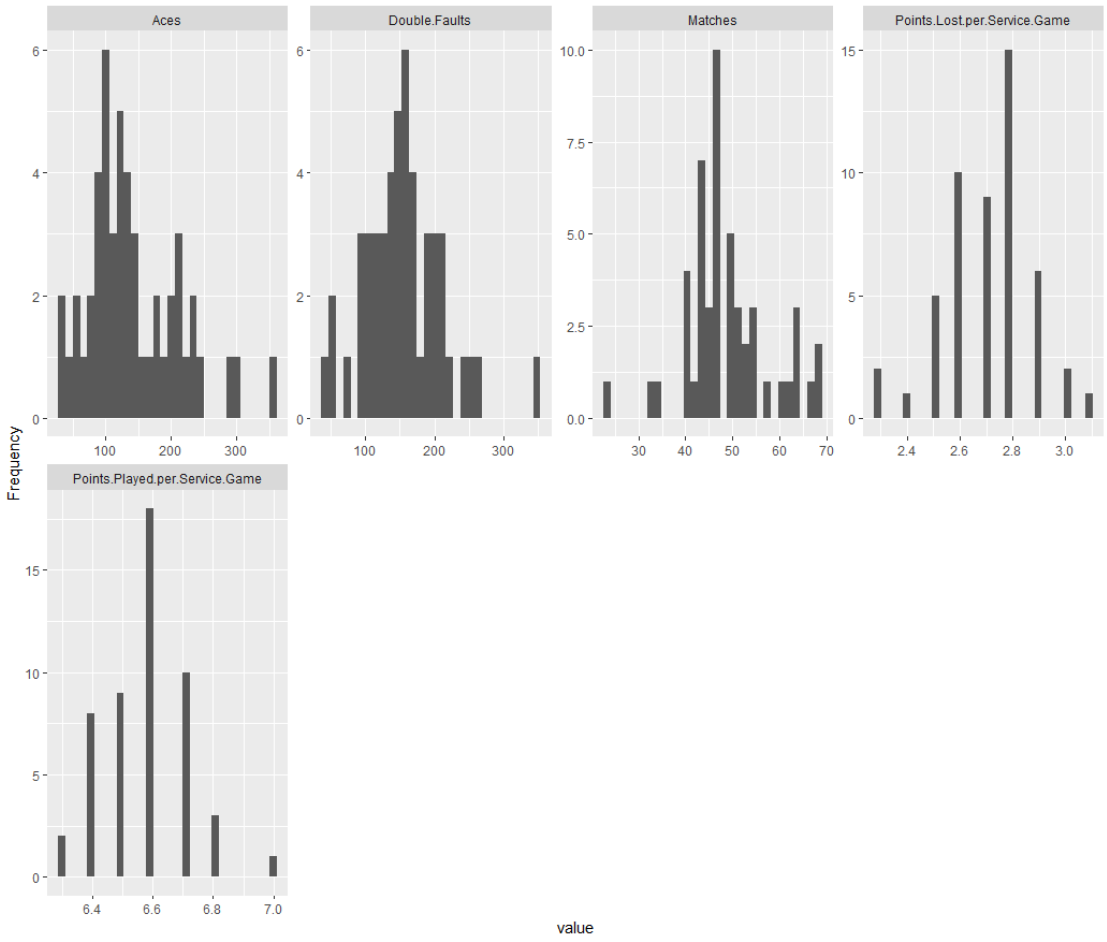


Figure 2 - Frequency Plots for Continuous Variables (Scraped Data)

From the above we see the frequency for each of the continuous parameters. However, it is not very clear, so we plot the density plot, below.

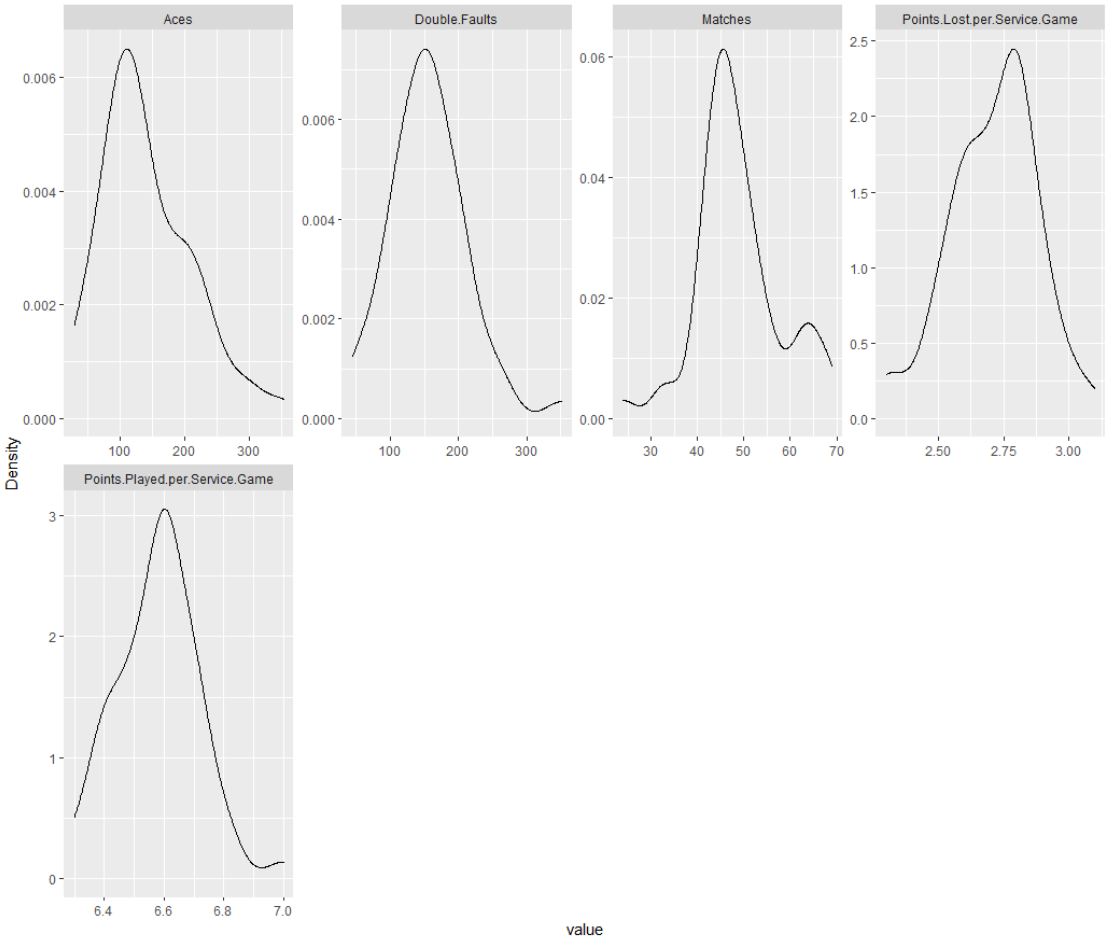


Figure 3 - Density Plots for Continuous Variables (Scraped Data)

From the density plots, we see the most common number of aces – around 110, double faults – around 165, matches – around 50, points/SG – 6.6, and points lost/SG – 2.80.

For the continuous variables from the Jeff Sackmann dataset, the exploratory plots seen below is for all the data that is available, not only the ones filtered for Grand Slams. This is because when filtered for Grand Slams, the number of observations dropped drastically. It was enough to

be used for the models to serve the original aim; but for EDA, the idea is to see as much of the data as possible to explore it. The first plot here is a frequency plot for the continuous variables.

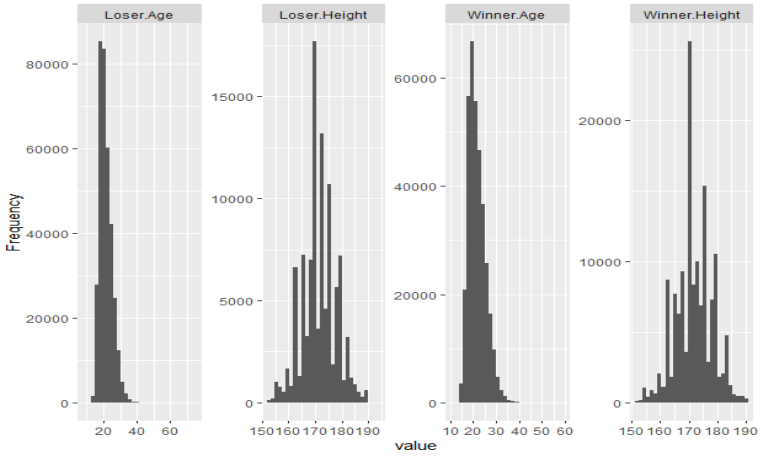


Figure 4 - Frequency Plots for Continuous Variables (Original Data)

Then, the density plot for better understanding is seen below.

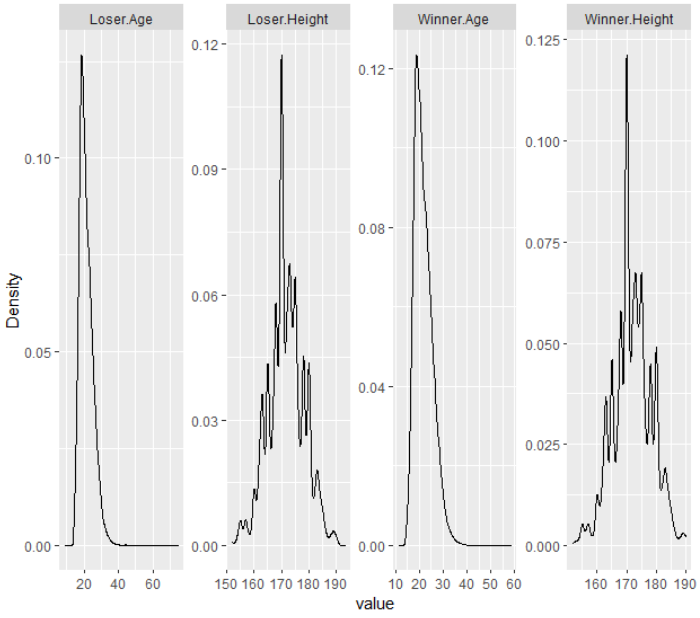


Figure 5 - Density Plots for Continuous Variables (Original Data)

The interesting insights from this is the age of the players that lose is mostly under 20; this hints that they are new professionals. The mode height for both losers and winners of the games are 170cm. The age of the winners is only around 22 years old (also young).

The next plots have to do with the exploration of different tournaments in the dataset. In this work, the focus as said was on Grand Slams where three different surfaces are studied. Roland Garros is played on clay surface, Wimbledon is played on grass surface, and the Australian and US Open are both played on hard surfaces. The below plot shows the count of the matches on different surfaces.

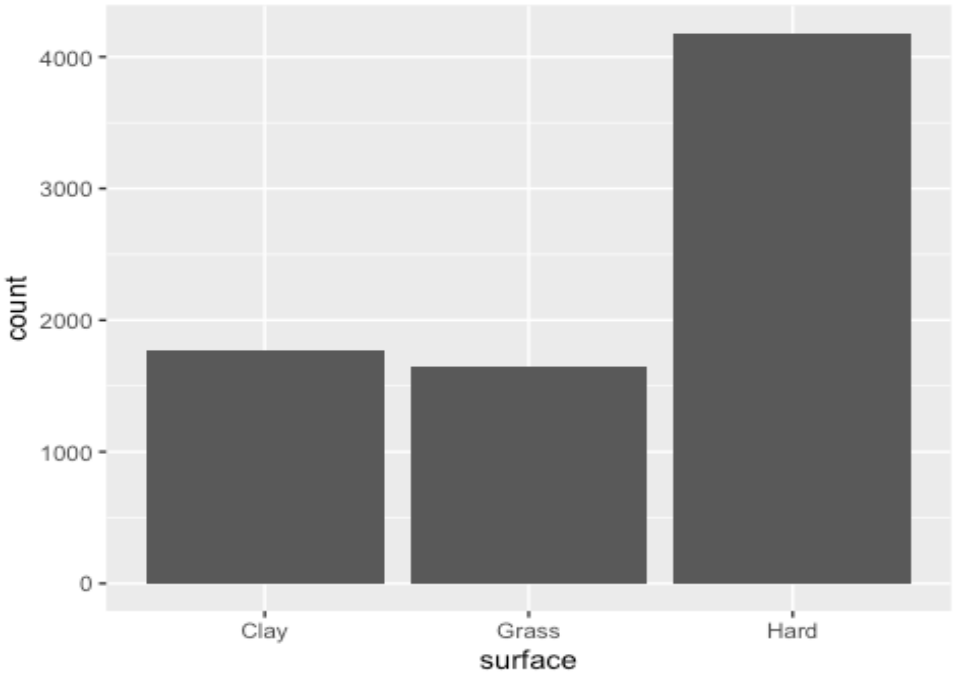


Figure 6 - Count of Matches Under Different Surfaces to Discover the Most Common Tournament in the Data

From the above plot, we can conclude that Wimbledon is the least played amongst the data used. We cannot conclude about hard and clay surfaced tournaments because two tournaments are played on the hard surface. The next plot aims to give more insights about the tournaments.

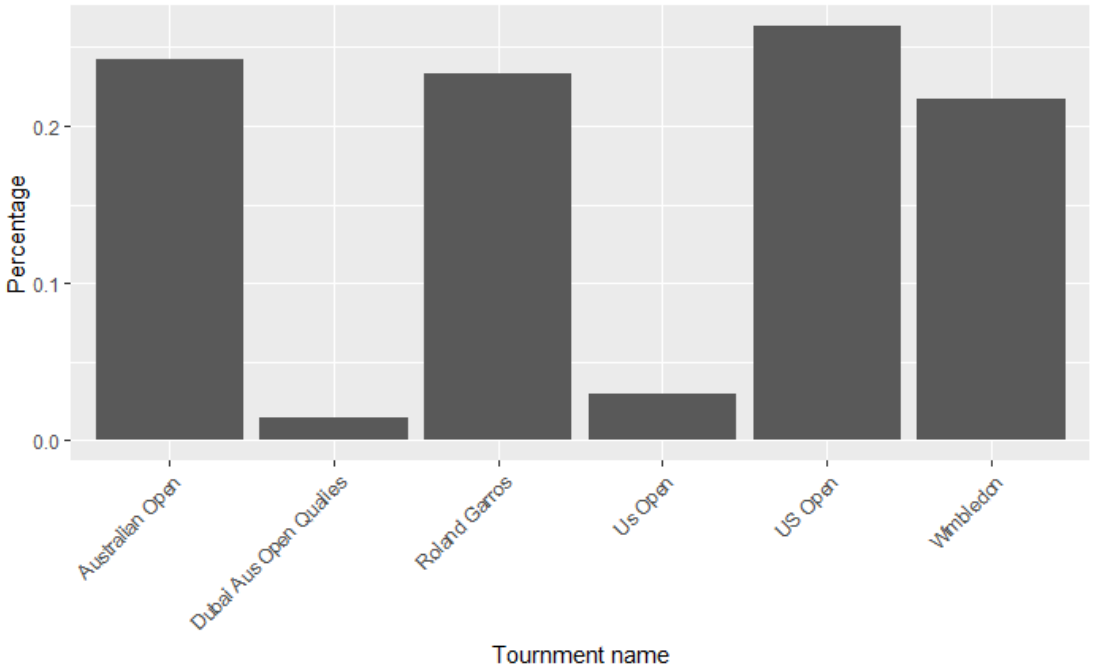


Figure 7 - Plot of Percentage Occurrence of Every Grand Slam Relative to the Other

This plot reveals some hidden information. It is seen that our data has two names for US Open and the qualifications of the Australian Open are documented, separately. This is was not seen in the analysis because it was enough to only filter for “Tourney_type” as ‘G’ which stands for Grand Slams.

The next plots that will be seen and explained answer some hypothesis to prove the objectives explained before. As mentioned, these

will be here since they were analyzed by exploring correlations in the data rather than applying methods, like the results in the next section.

Starting off with the first objective of understanding how anthropometric data affects the performance and results of players, the data is filtered for players in Grand Slams and players that have played over 40 matches.

The first plot seen below explores whether changes in heights matter. It is a simple plot of the distribution of the height difference between opponents.

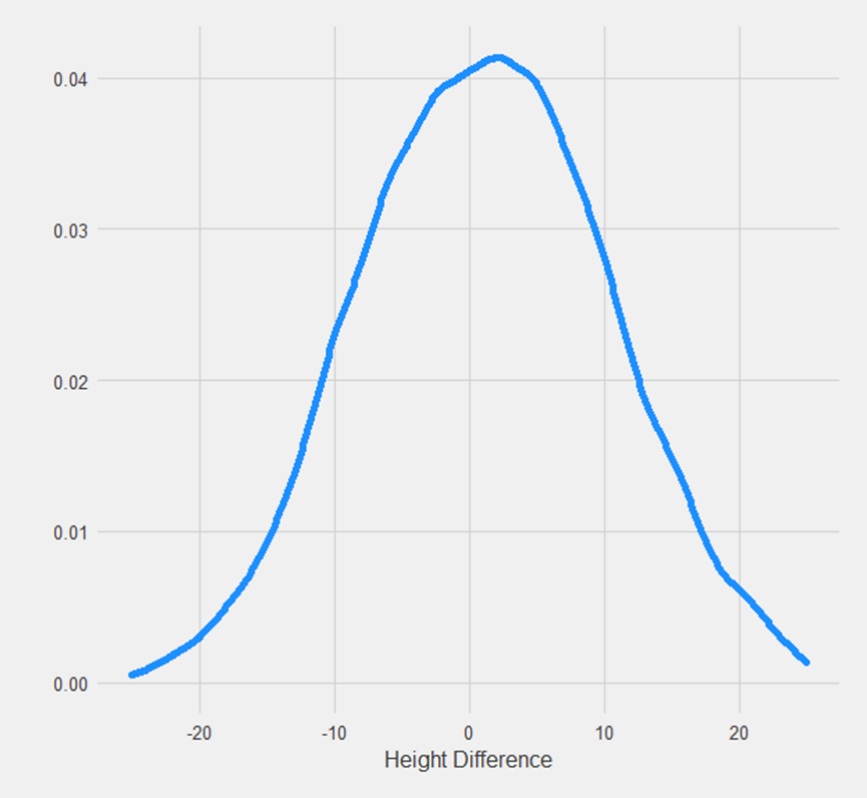


Figure 8 – Distribution Plot of Height Differences Between Winners and Losers

As close as the distribution is on top of 0, it is not perfect. That suggests that there is some height difference between players that win and players that lose. We could also take a look at the distribution of heights this time.

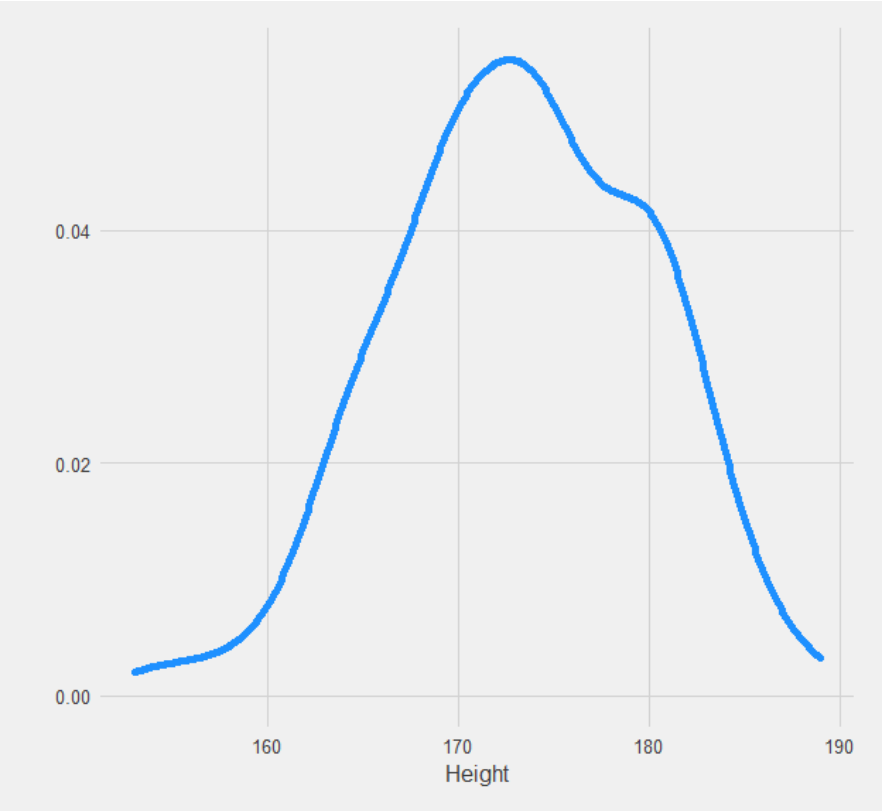


Figure 9 – Distribution Plot for Heights of all Players

We see that there is clearly a difference in height for different players so the mission is to understand about this difference and how it affects the play. This is done with some bivariate analysis and, first, there is a plot for the number of aces against different heights of players to see if it is possible to see any difference for short versus tall players.

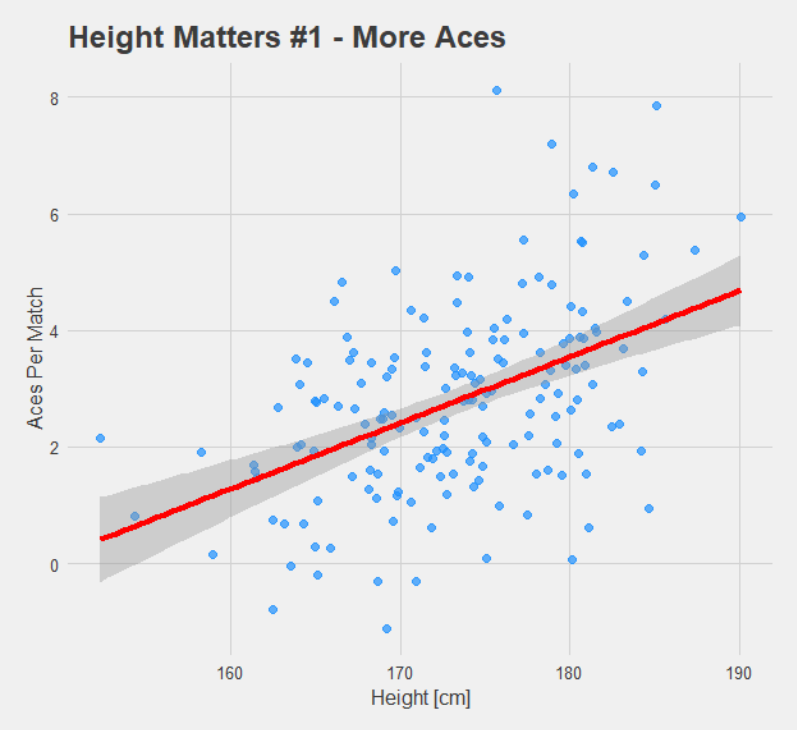


Figure 10 – Scatter Plot of Aces per Match Against Height - red line is a line of best fit, the blue dots represent aces per match for various players

From the above, it is seen that the number of aces per match increases for taller players, indeed. This plot explains some sort of performance affected by anthropometric data but it does not quite explain if height helps players win, so we do a plot of the different ranks for different heights and this is seen below.

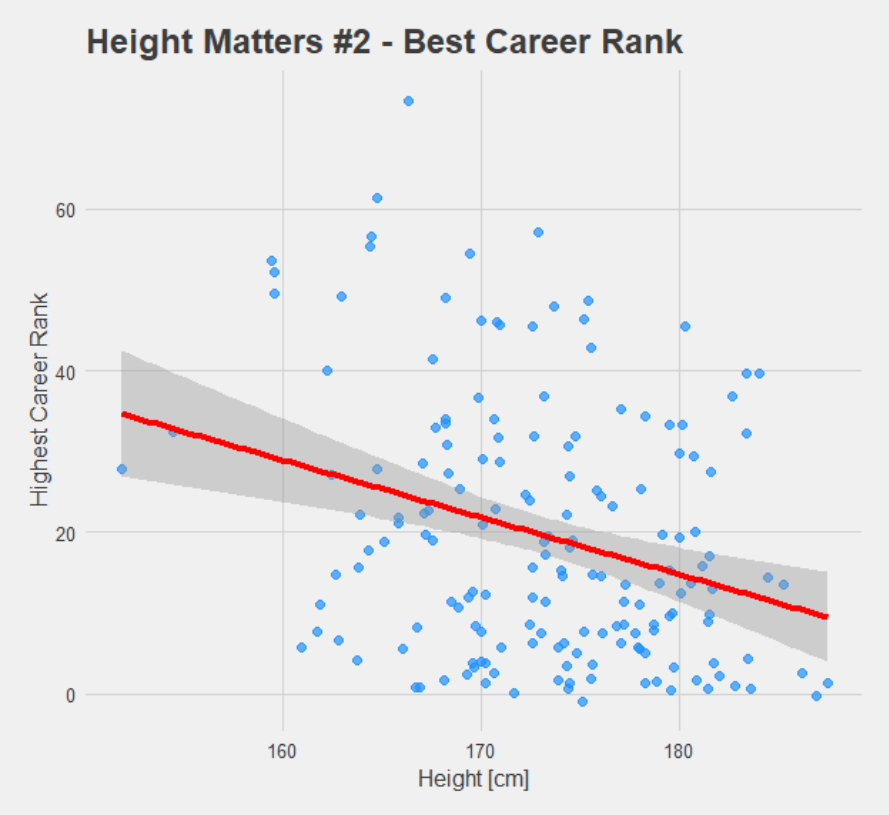


Figure 11 – Scatter Plot for Players' Best Rank Against Height – the red line is a line of best fit, the blue dots represent highest rank for various players.

The above plot considered, for every player, the best rank she has ever reached and it is clear that the lower – better – ranks have been occupied by taller, rather than shorter players.

Since it is clear that ranking is affected by height, the probability of winning should be explored. First, we see the general winning percentages for different heights in the below plot.

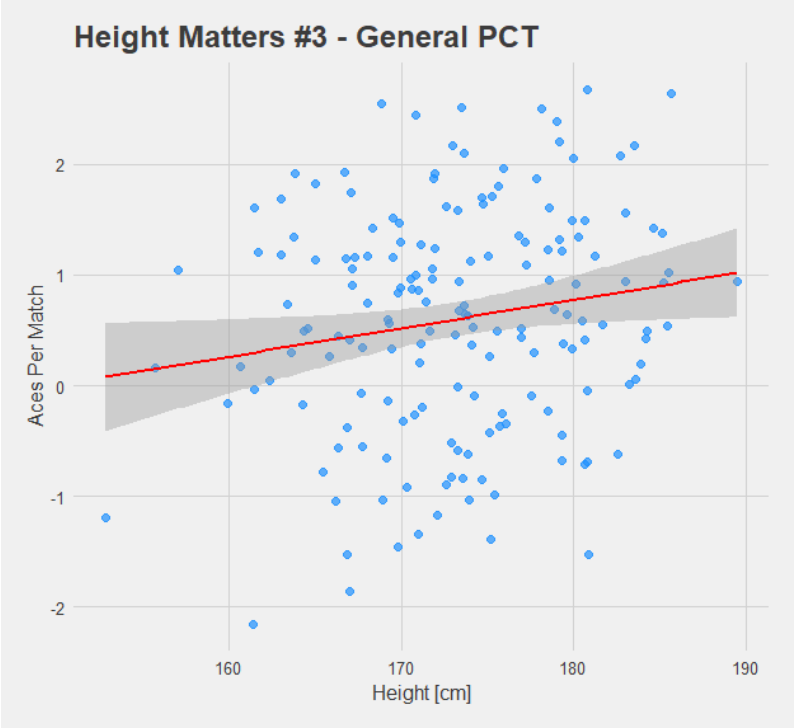


Figure 12 – Scatter Plot of General Winning Percentage Against Height – the red line is a line of best fit, the blue dots represent the aces per match for various players

To get more details, the next plot shows the probability of winning with every increase in one centimeter.

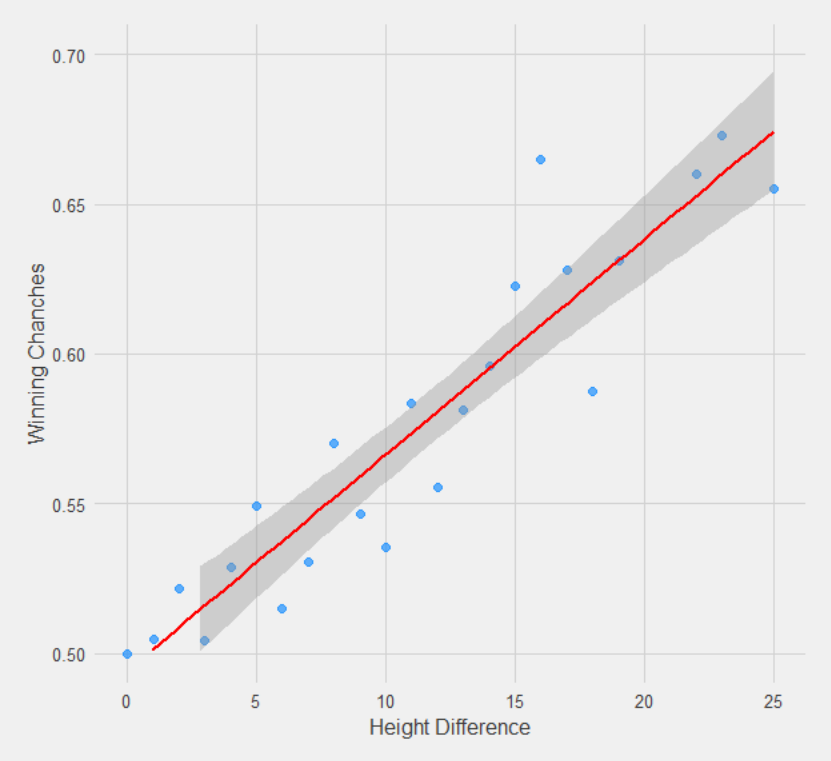


Figure 13 – Scatter Plot of Winning Chance Against Height – the red line is a line of best fit, the blue dots represent the winning chances for various players

The plot indeed proves that there is a better chance of increase with every centimeter of extra height from player to another. This brings up the curious question of whether heights are varying over the years since tennis is improving and becoming more competitive over the years. The below plot shows the evolution of height over the years.

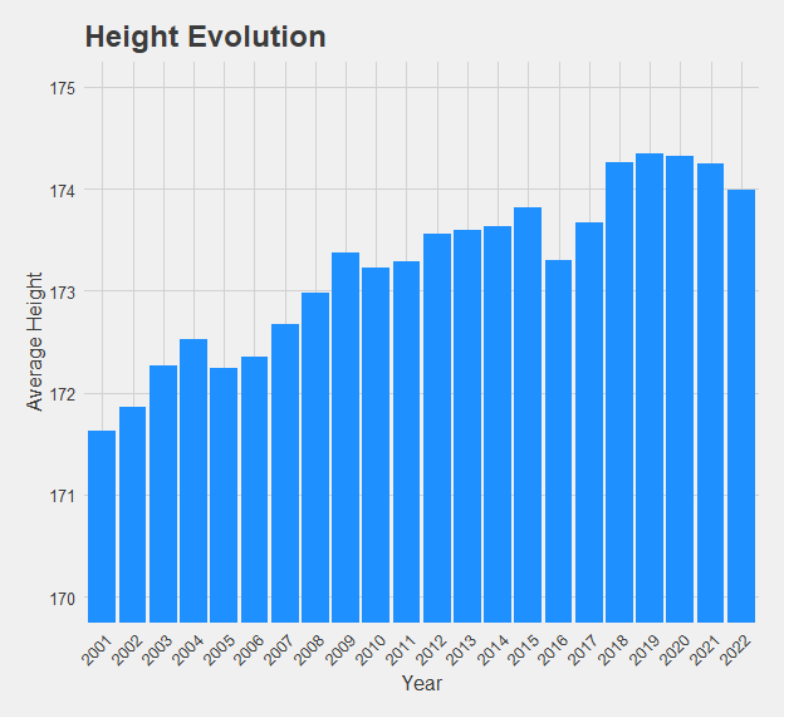


Figure 14 – Histogram for Height Evolution of Players' Average Heights over the Years

Indeed, height of players is increasing over the years of analysis. Height might as well be a factor for the improvement of tennis over the years. This is further supported with the statistical result of the average height of players versus the average height of best ranked players. It is not surprising that the obtained results are 173.27cm and 177.26cm. There is around 4cm difference between them! Now, moving on to age, let us see the probability of win for different age groups.

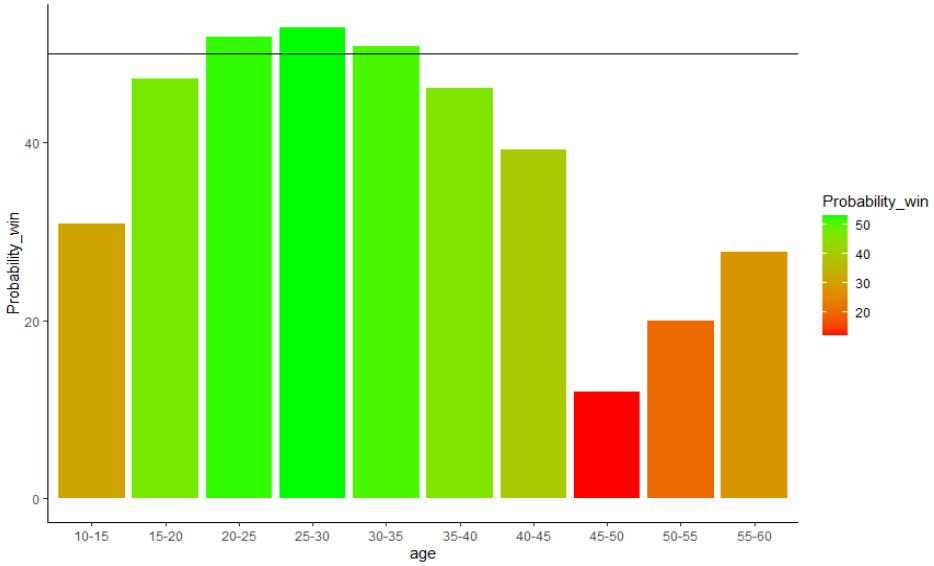


Figure 15 – Histogram of the Win Probability for Different Age Groups

We can see that the obvious groups from ages 20 years to 35 years have the most winning probability, especially the group of 25 years to 30 years. To get more detail we plot the age groups probability of winning when they face off using a correlation plot for the probabilities.

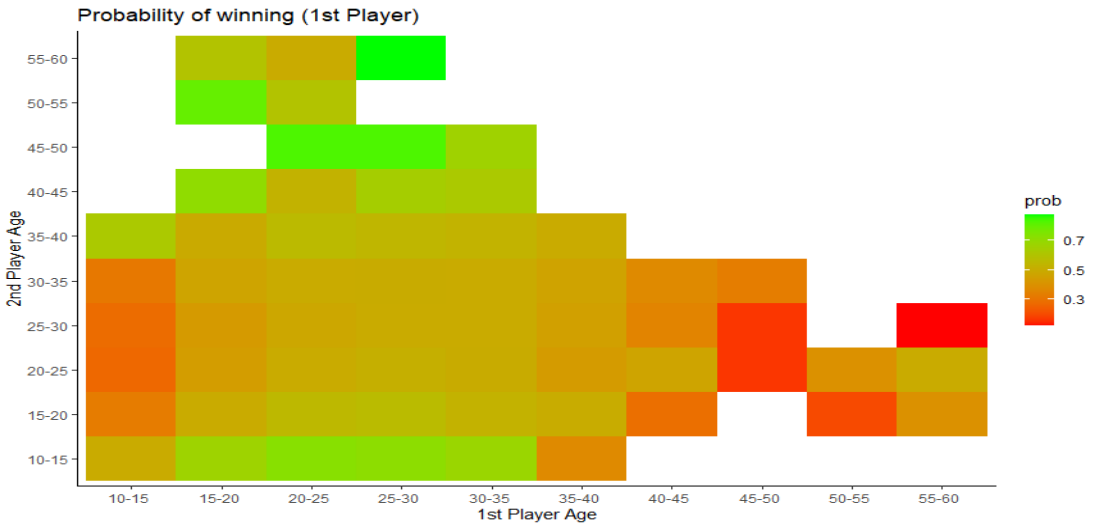


Figure 16 – Correlation Plot for the Win Probability for Age Group Face-offs

We cannot say there are any surprising results from the above, but it is interesting to see that our commonly thought hypotheses are correct. The next plots are added here because they are showing age advantage in the last set and in the finals. However, initially, they were done when analyzing the third objective which has to do with sport psychology factors affecting performance due to pressure of nerves. So here, we are trying to see if older players do better in the last set from the below plot.

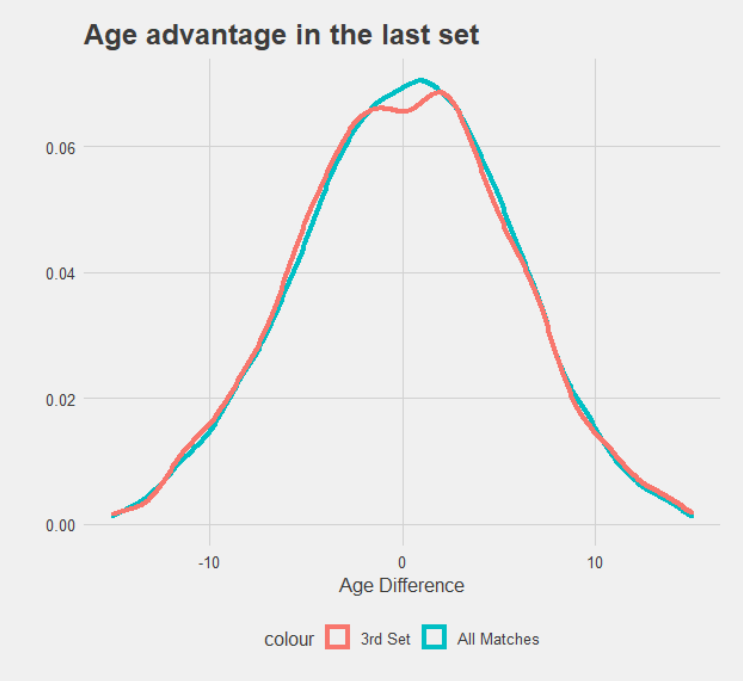


Figure 17 – Distribution Plots to Show Age Advantage in the Last Set (red) and All Matches (blue)

The hypothesis was that older age helps in the last set and optimally, the results would show the histogram more skewed towards older players in the last set but the difference is not that much. We cannot

forget that age is not the only factor that helps to win or lose in a match even if it was the third set. In the next section, the results of the applied Kolmogorov-Smirnov test to check for statistical distribution differences are seen.

But let us see if the results are better for age advantage in the finals from the next plot.

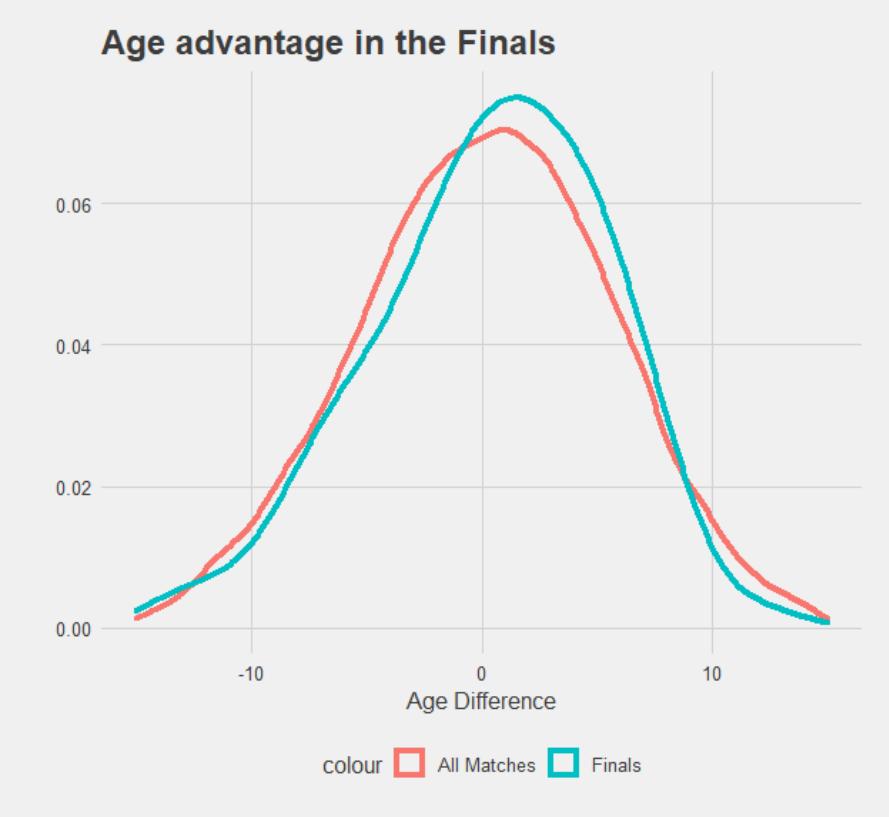


Figure 18 - Distribution Plots to Show Age Advantage in the Finals (blue) and All Matches (red)

Ignoring the limitations from the previous plot and the fact that there are not as much data about finals as other matches, we can see that

older players do better in finals than in the third sets. It could be that the finals are considered more stressful than general three sets but this is open to different interpretations. However, this cannot be concluded visually; the two distributions look different but they may not be statistically different. The results of the Kolmogorov-Smirnov test on the distributions is seen in the next section.

Additionally, since there was a lot of curiosity in the age effects, it was important to note that experience is what is forming the hypotheses and experience is not only with old age. In other words, some players might start off old and have less experience than players at that age who started younger. The next plot shows the optimal years of play to reach finals, in other words, the optimal years of play for advantage due to experience.

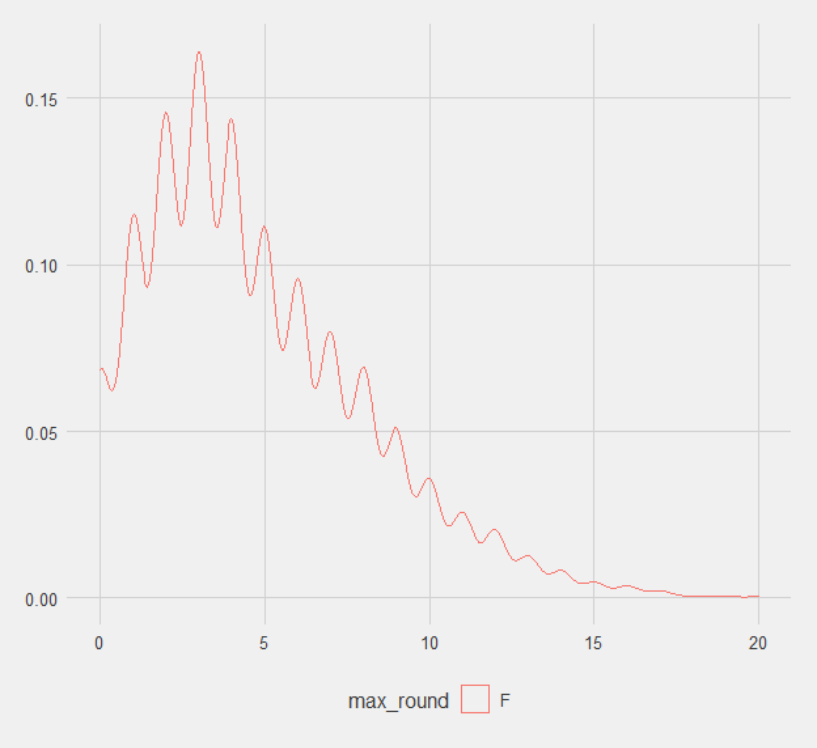


Figure 19 – Density Plot for Reaching the Finals Against the Years of Experience

From the above plot of geometric density against years of experience, it is seen that the optimal years of experience are around 4 years, regardless of the starting age.

Now, the focus switches to the second objective of predicting results and performance patterns from match data, specifically different surfaces. In tennis, there are four different kinds of surfaces, namely: clay, grass, carpet, and hard surface. In the Grand Slams, none of the four tournaments is played on carpet. This is why all the analysis below shows only three types of surfaces. The first plot shows the number of aces per match on different surfaces.

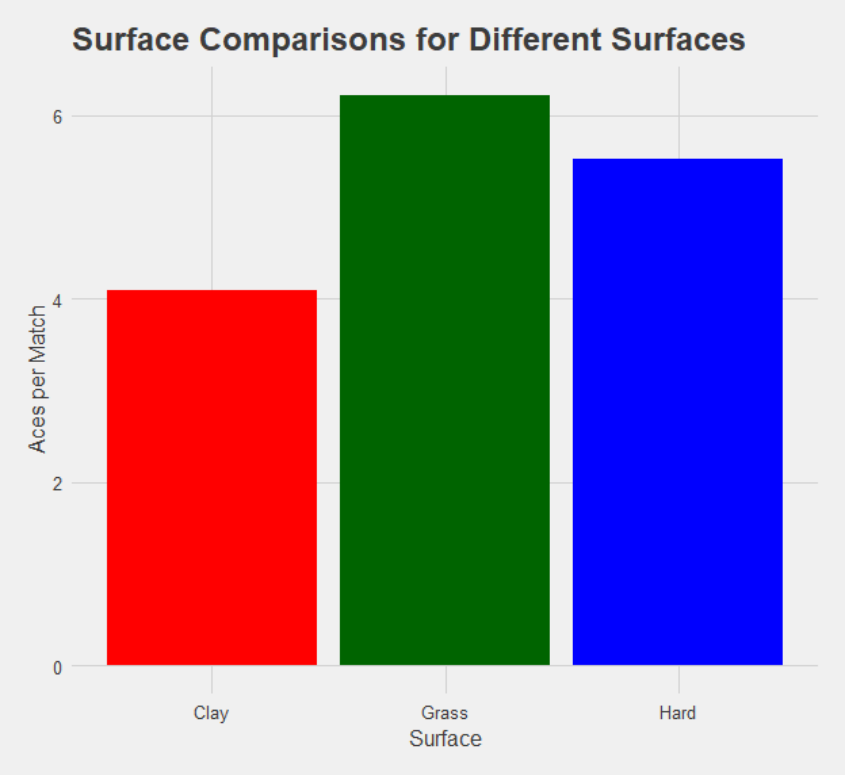


Figure 20 – Bar Plot for Aces per Match on Different Surfaces

It makes sense that the number of aces is significantly lower on clay ground. This is because clay surfaces are covered with sand which reduce the speed of the ball and hence, make it possible for the opponent to get the serve and not concede an ace. This is different for harder surfaces with less friction reducing the speed of the ball. The next plot shows the aces per serve on different surfaces.

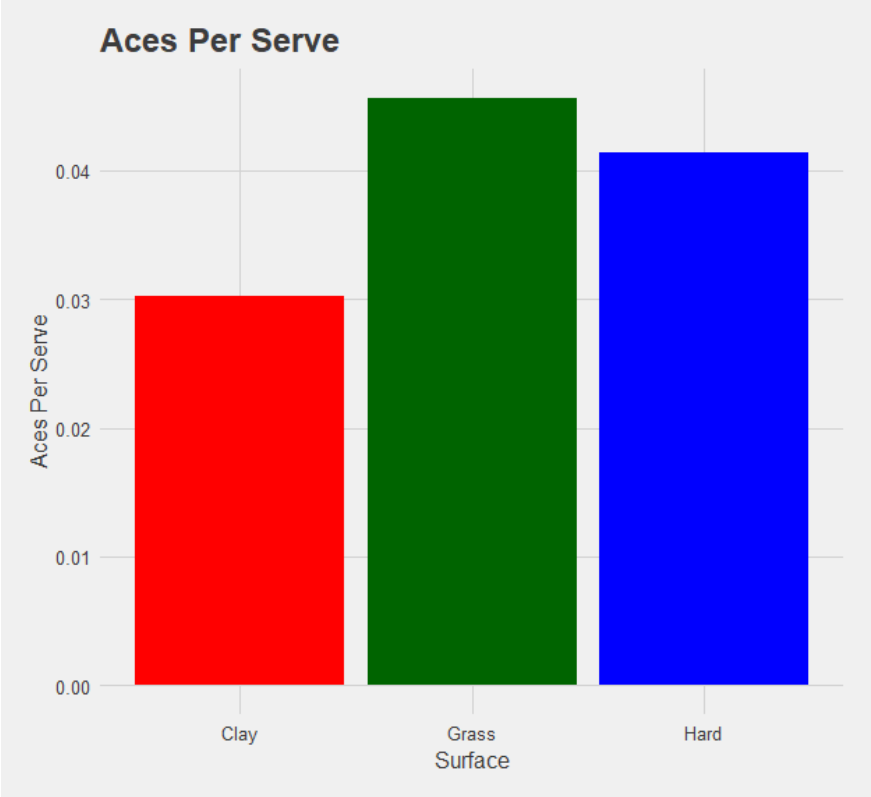


Figure 21 – Bar Plot for Aces per Serve on Different Surfaces

Again, the numbers are lower for clay surfaces. This proves the hypothesis of having less aces per serve on a slower surface for the same reasoning of numbers of ace per match.

The next plot aims to look at the percentage of first serves on different surfaces.

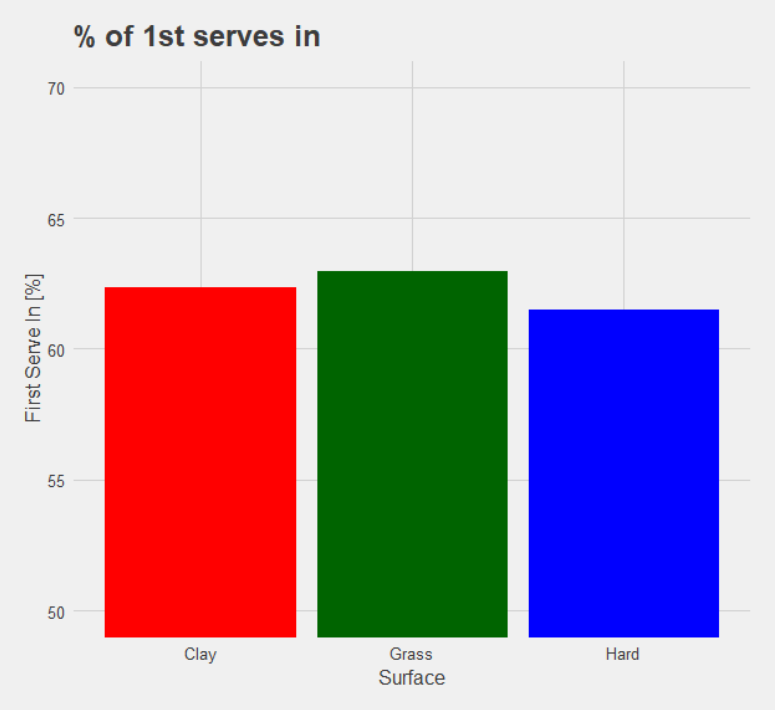


Figure 22 – Bar Plot for First Serve Percentage on Different Surfaces

Unfortunately, this did not give interesting insights. This is probably because not only the speed of the ball is affected on clay surface but also the speed of the player. This compensates and hence gives similar percentages for every surface. Looking further into this, the next plot shows the percentage of first serve points taken on different surfaces.

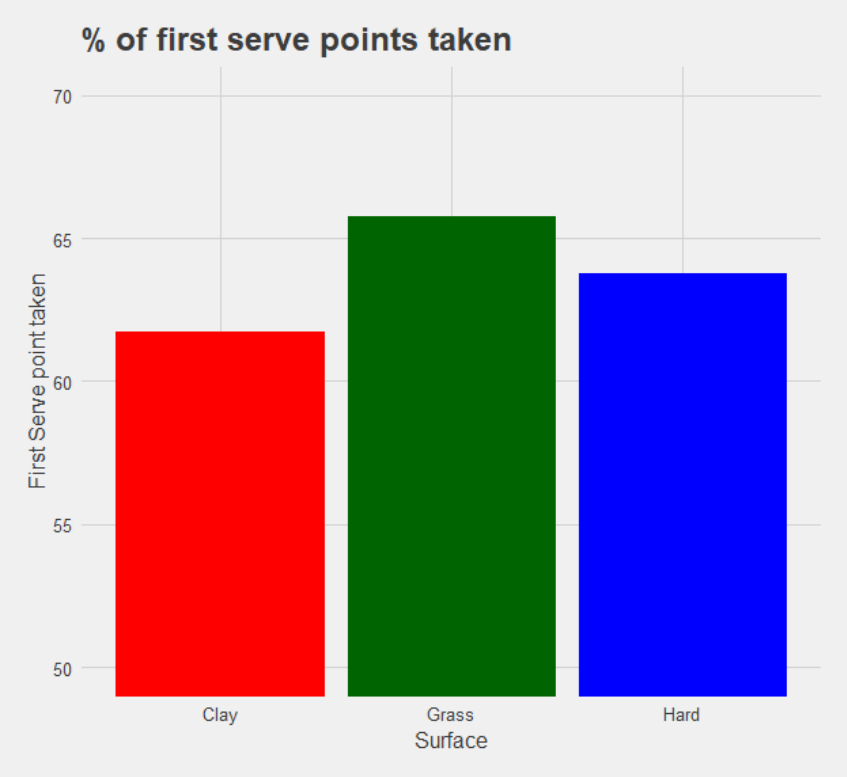


Figure 23 – Bar Plot for First Serve Points Taken on Different Surfaces

When matches are played on clay surface, in the Roland Garros tournament specifically, we notice that the rallies are longer. That means that we have more exchange per play for every serve. This may explain the above results. In clay surfaces, there are less first serve points taken because the longer rally exchange reduces the odds for winning due to serve advantage. A further investigation in future work could look into break points and we hypothesize that there are more break points on clay surfaces.

Last part of exploratory data analysis focuses on some descriptive analysis for the third objectives which has to do with sport psychology. In here, there are analysis for two things: upsets and comebacks as explained before. Starting with upsets, let us understand when do upsets happen. This can be seen in the plot below that compares winning probabilities for rank differences.

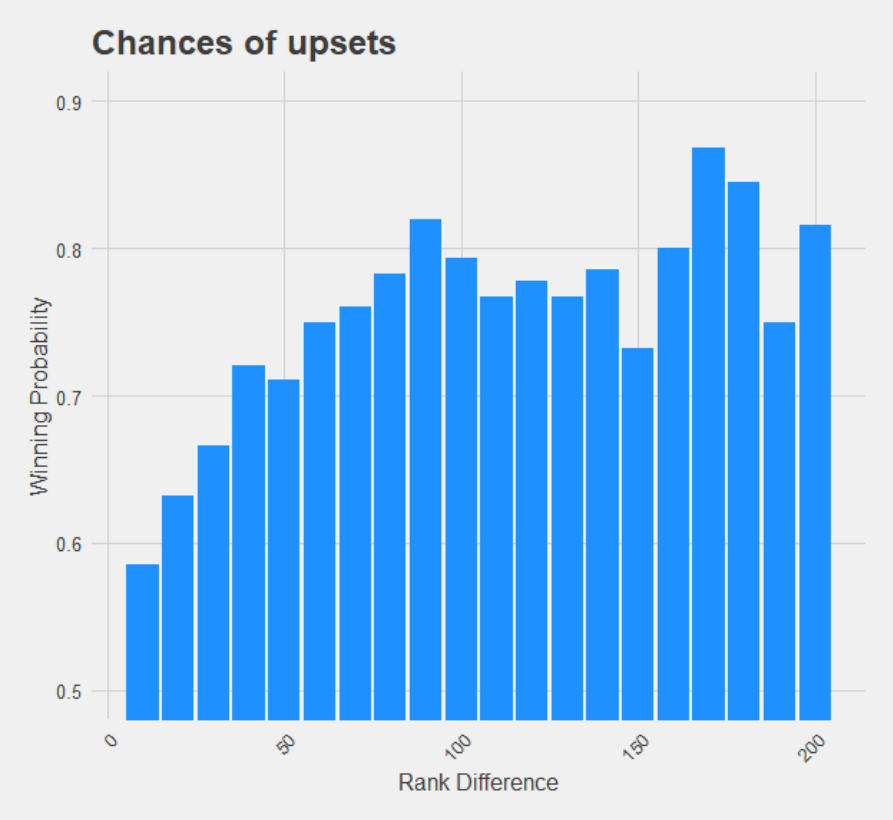


Figure 24 – Histograms of Winning Probability against Rank Differences – this is to show upsets happening for various rank differences

The above plot aims to show that upsets happen mostly when there is a bigger different in rank. From the above plot, we see that when there

is a bigger rank difference, the favorite with better rank has more probability of winning. Therefore, upsets happen when there is not that much of a competitive advantage for the player with better rank.

Next, we try to see if this changes from one surface to another.

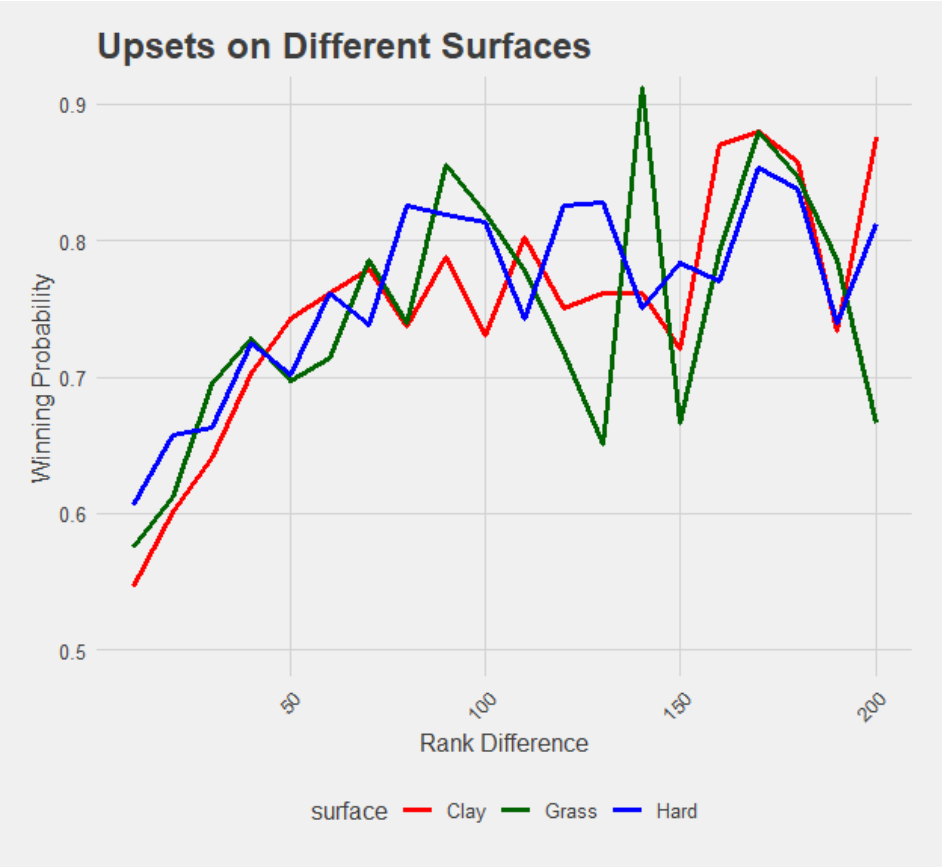


Figure 25 – Plot of Winning Probability Against Rank Differences for Difference Surfaces – this is to show upsets happening on different surfaces

The surface does not seem to affect the upset phenomena that much. There is a similar trend for winning probabilities and rank differences on different surfaces.

Let us check if this changes when comparing the occurrence of upsets in different tournaments. The next plot shows how upset differs in big tournaments (Grand Slams) and in smaller tournaments.

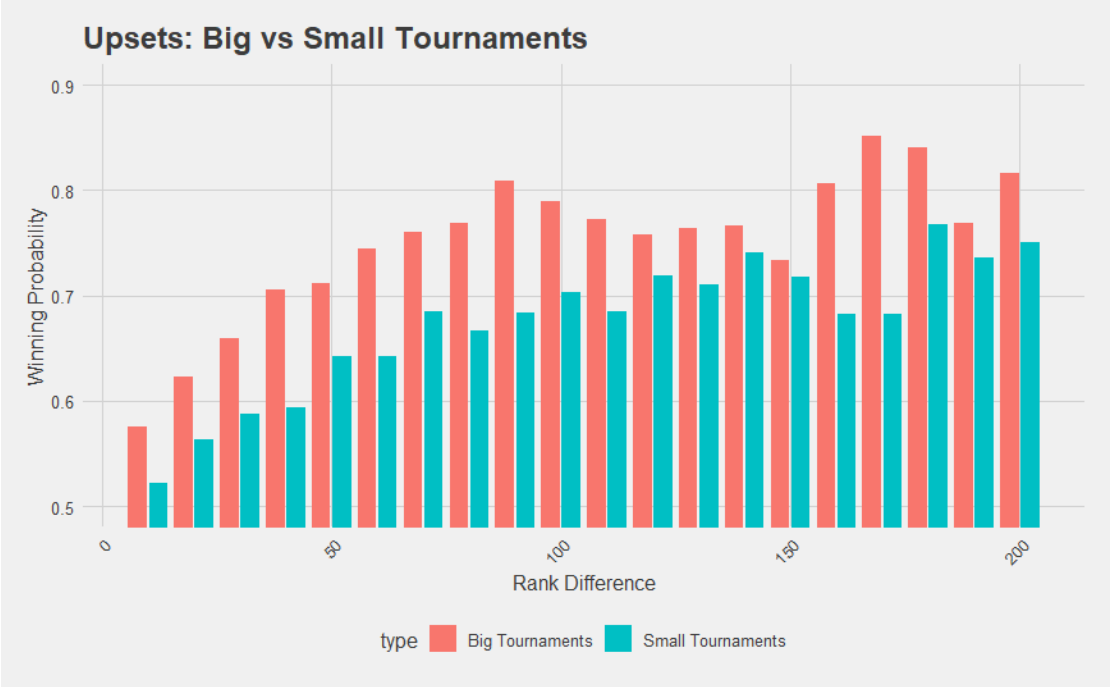


Figure 26 – Histograms for Winning Probability of Different Ranks – this is to show upsets happening in big tournaments (red) and small tournaments (blue)

We can see that the occurrence of upsets differs from one tournament type to the other. The above figure explains that it is more probable for a better ranked player to win in a big tournament. It makes sense that players take the smaller tournaments more lightly than bigger ones and therefore, it is more difficult to beat a better opponent – perform an upset – in a Grand Slam. This could be due to better mental and physical preparations in big tournaments for all players in general, especially the big

players that know they will earn more if they do better in bigger tournaments. Another thing is that in a bigger tournament, the atmosphere is probably more stressful but there are less skilled and experienced players in there that can perform an upset; this is because the number of players in bigger tournaments is larger.

Now moving on to the second part of the third objective, let us get some insights about comebacks. The first thing to look at is the statistical result of the percentage of comebacks occurrences and the result is 15.83%. That is to say that when a player has won her first set, she already has a winning probability of 85%!

Then finally, we try to look at the type of players that make comebacks in the next plot.

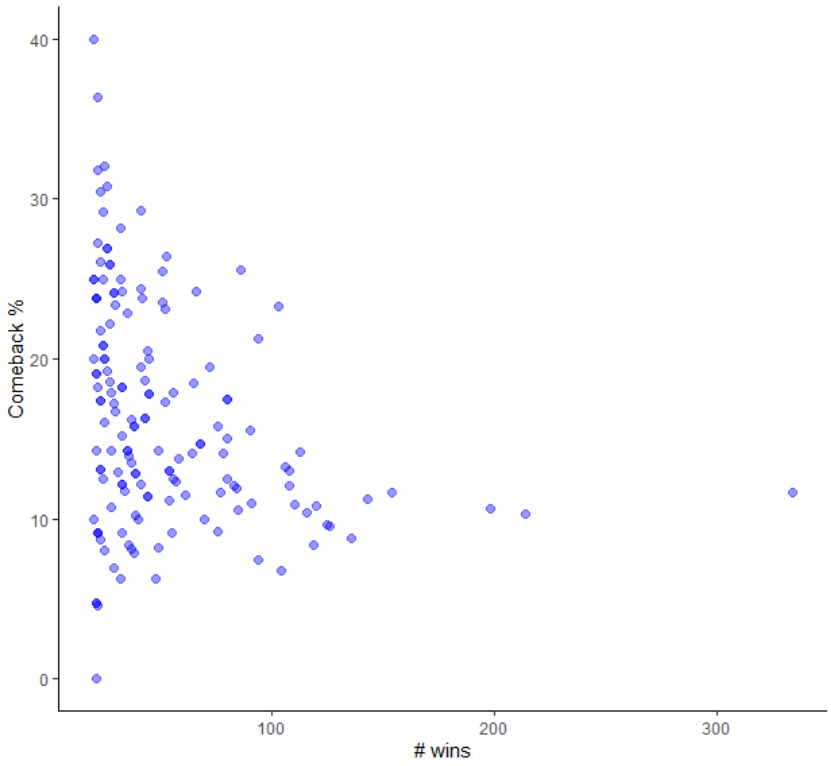


Figure 27 – Scatter Plot for Comeback Percentage Against Number of Wins for Various Players

The above plot shows that players with a smaller number of wins, make the most comebacks. This makes sense because better players, or those that have a higher number of wins, comfortably win in comparison to worse players that may need to turn things around after losing the first set. It is interesting to see also that as the number of wins increases, the percentage of comebacks moves to the average.

3.2. RESULTS OF STATISTICAL METHODS APPLIED

The results of the rest of the analysis will fall here before discussing and concluding in the last chapter. Starting with the statistical analysis for the results of exploratory data analysis for surfaces, Kruskal-Wallis test and its effect size were computed to see if there is any statistical difference between each metric used and the three different surfaces. Before that, an Analysis of Variance model was done and all the p-values less were less than 0.05; now we check the Kruskal Wallis test.

Metric per 3 Different Surfaces	Chi-Squared	Degrees of Freedom	p-value obtained from Kruskal Wallis Test	Effect Size Magnitude
Aces per Match	17149	2	< 2.2e-16	Large
First Serve Rate	8572	2	< 2.2e-16	Large
Rate of First Serve Points Taken	383.4	2	< 2.2e-16	Large
Aces per Serve	8572	2	< 2.2e-16	Large

Table 3 - Tabular Results of Kruskal Wallis Test and Magnitude of Effect Size for Every Metric per Surface

From the above results, we see that there are significant differences for every metric on different surfaces because the effect size magnitudes are large and the p-values of every metric are a lot less than 0.05.

Next, we take a look at the results of the Kolmogorov-Smirnov test to check for statistical distribution differences in the two distributions of the plots for the advantages of age in the finals and ages respectively.

Distributions Tested	p-value
Age advantage in the last set	0.5961
Age advantage in the finals	0.9251

Table 4 - P-value Results of Kolmogorov Test to Check for Statistical Differences in Distributions

The above results of the Kolmogorov Smirnov test give a p-value greater than 0.05, in both cases. This tells us that the distributions are not different. In other words, it is not possible to conclude that there is an age advantage in the third set, or in the finals.

Moving on to models, we look at the results of logistic modelling for the third objective where the goal was to understand the effects of mental resilience. In this plot the aim is to see how many sets are needed to perform an upset. Does a player that is beating a better player require

just two sets to beat her or would she require the full three sets of the match?

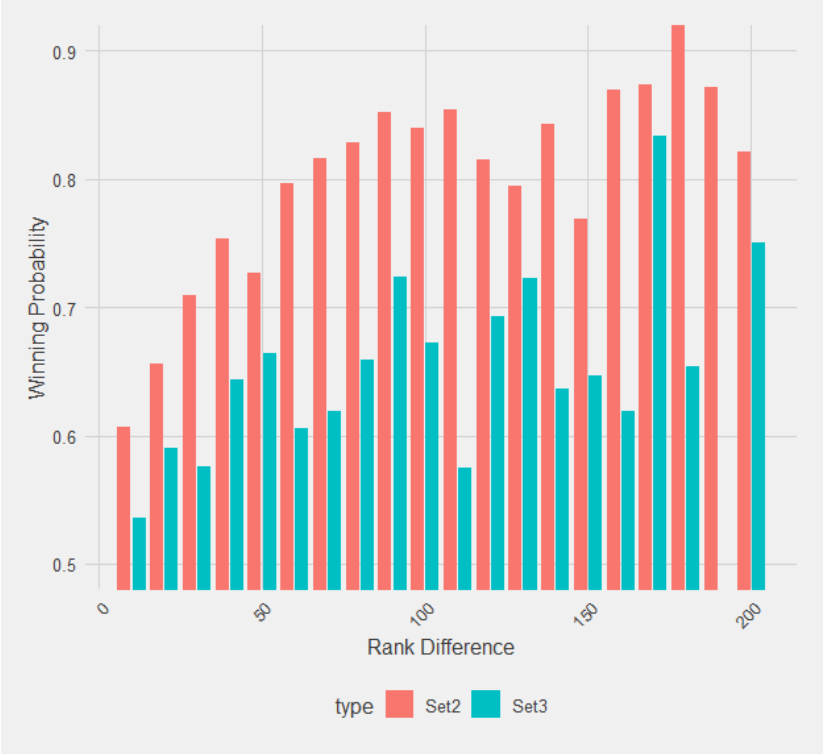


Figure 28 – Histograms for Winning Probability in Different Sets – this is to show upsets happening in two sets (red) and three sets (blue)

The results show that a player that needs to perform and upset generally needs three sets. We see that better players are more comfortable on two sets.

Moving on in the surface analysis, 6 clusters were created to see the winning percentages there. This is done with K-means clustering based on winning percentage for each group of players and seeing the centers to tell us something interesting. Only 6 clusters were used so that we have

them grouped on general level (winning probability). Also, the average height of players is added for every group.

HardPCT	GrassPCT	ClayPCT	Avg Height (cm)
0.763	0.753	0.775	177
0.379	0.401	0.457	172.77
0.659	0.658	0.628	174.66
0.533	0.503	0.598	173.72
0.472	0.536	0.327	170.82
0.335	0.240	0.257	170.95

Table 5 – Winning Percentage Clusters

Then the clusters were plotted for better understanding.

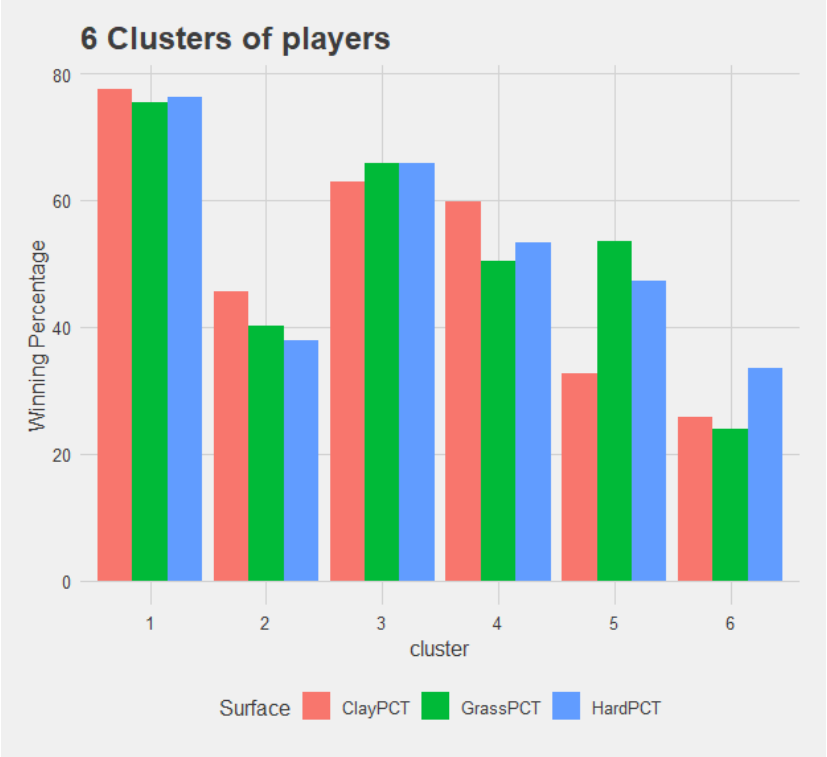


Figure 29 – Bar Plots of Win Percentage for Every Cluster

We see the players in cluster 1 and 3 have the highest winning percentage. For cluster 6 that has the lowest percentages, the probability is highest at hard surface. Moving on, the PCA of the clusters is plotted below.

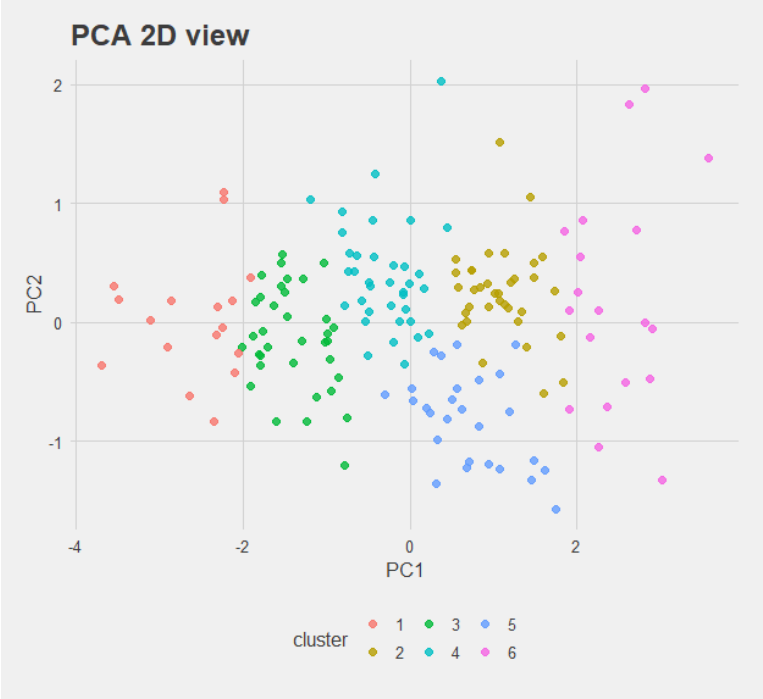


Figure 30 - PCA Plot of Clusters

For models, the first thing is the logistic regression model that was done. The logistic regression model tries to predict if the best scored in terms of points always wins, so to do that, the independent variable considered was

X = winner_rank_point - loser_rank_point

and the dependent variable as

y = int (X>0)

This means “y” will be 1 or 0 (TRUE or FALSE). In other words, if y = TRUE, X > 0 meaning that winner_rank_point is greater than loser_rank_point. Basically, if y = 1, the winner has more points than the

loser and if y = 0, the winner has less points than the loser. The data is split in two for training and for testing the model.

Below is the summary of the model:

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.0151	-1.2473	0.7152	0.7964	0.9819

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	1.2484371	0.2998701	4.163	3.14e-05
surfaceGrass	-0.1408914	0.3165744	-0.445	0.6563
surfaceHard	-0.0016782	0.2762947	-0.006	0.9952
diff_seed	-0.0009453	0.0168917	-0.056	0.9554
age_diff	0.0254060	0.0203769	1.247	0.2125
ht_diff	0.0041964	0.0098025	0.428	0.6686
rank_diff	0.0026666	0.0093013	0.287	0.7743
experience	-0.0541870	0.0288995	-1.875	0.0608
last_resultQ2	0.1489365	0.2395450	-0.622	0.5341

last_resultQ3	0.5440826	0.2608323	2.086	0.0370

Table 6 - Results of Logistic Regression Model

```
Null deviance: 592.51 on 521 degrees of freedom
Residual deviance: 580.77 on 512 degrees of freedom
AIC: 600.77
Number of Fisher Scoring iterations: 4
```

Next are the confusion matrix results where the model was tested on the test data.

Prediction	-1	1
-1	0	0
1	33	97

Table 7 - Confusion Matrix Results for Logistic Regression Mode

The accuracy obtained is 74.62%. At 95% confidence interval, the lower and upper bounds are 66.24% and 81.84%, respectively.

Then for feature importance, the results are seen below.

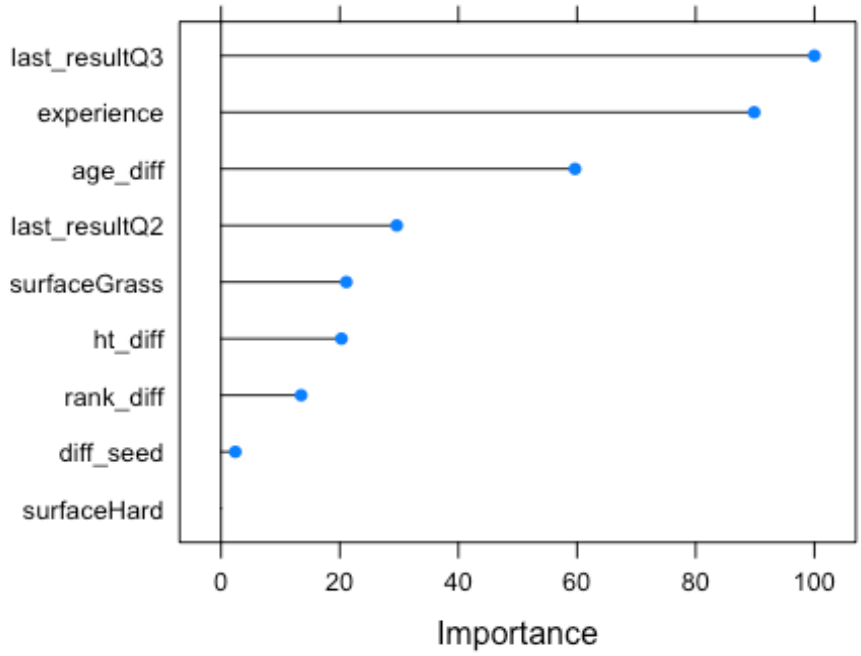


Figure 31 – Feature Importance from Logistic Regression Model

The summary of the logistic regression reveals that only experience and last_result covariables are statistically significant at 0.1 and 0.05 levels respectively.

Next, decision trees were applied to the data frame with the selected columns: surface, round, tourney_level, diff_seed, age_diff, ht_diff, rank_diff, experience, last_result, and win_lose. The result is below:

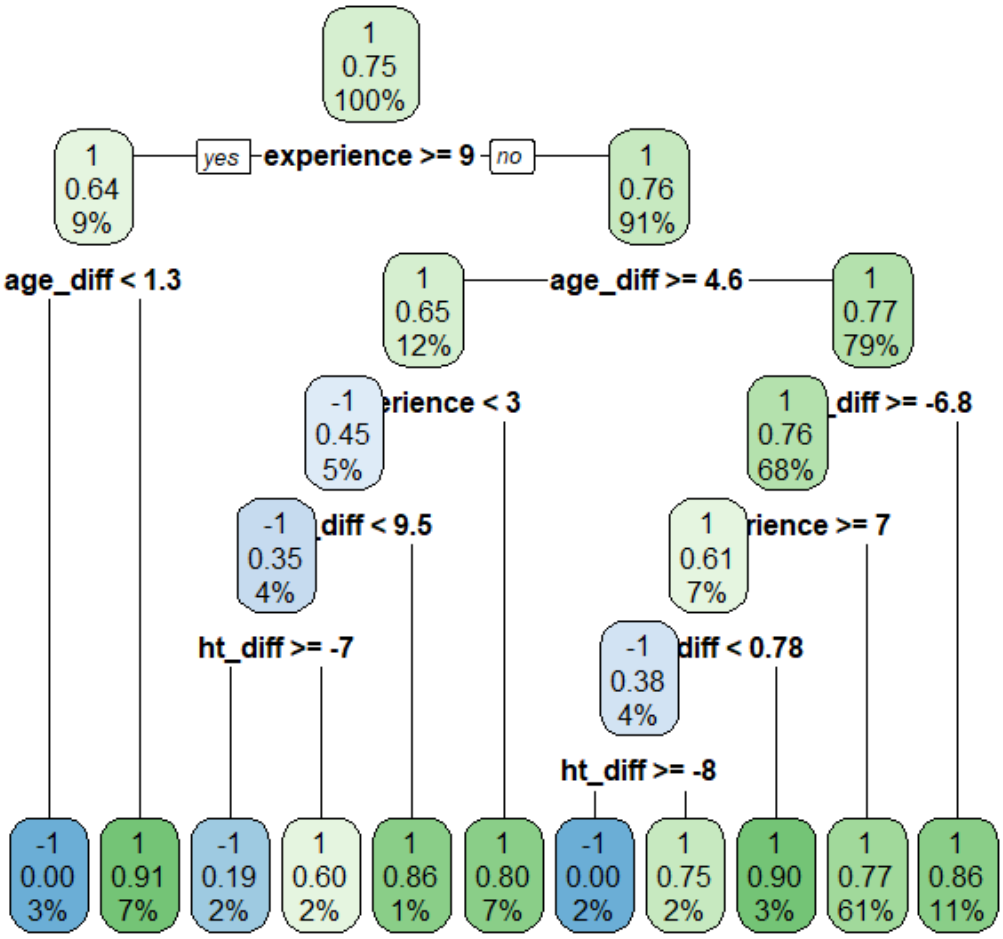


Figure 32 - Decision Tree Results

The tree is based on important variables only: the first node is based on the experience. If experience is less than 9 then we check for age, whether age is < 0.86, the prediction for the *win/lose* variable will be 1, otherwise -1 is predicted. On the other hand, where experience is greater than or equal to 9, -1 for the *win/lose* is predicted when age difference is less than 4.6 and experience is greater than or equal to 3 and height difference is greater than -3 or when age difference is greater than or equal

to 4.6 and last result is different from Q2 and random difference is less than 39. The other cases are predicted 1.

A good way to see the performance of a model is to calculate the confusion matrix.

Prediction	-1	1
-1	1	1
1	32	96

Table 8 - Confusion Matrix Results of Decision Tree Model

The accuracy obtained is 74.62%. At 95% confidence interval, the lower and upper bounds are 66.24% and 81.84%, respectively.

After the decision tree, a Random Forest model is done. This is to benefit from the added randomness and instead of focusing on the most important feature, we get to search for the best feature among a random subset of features. The results for the random forest model are seen below.

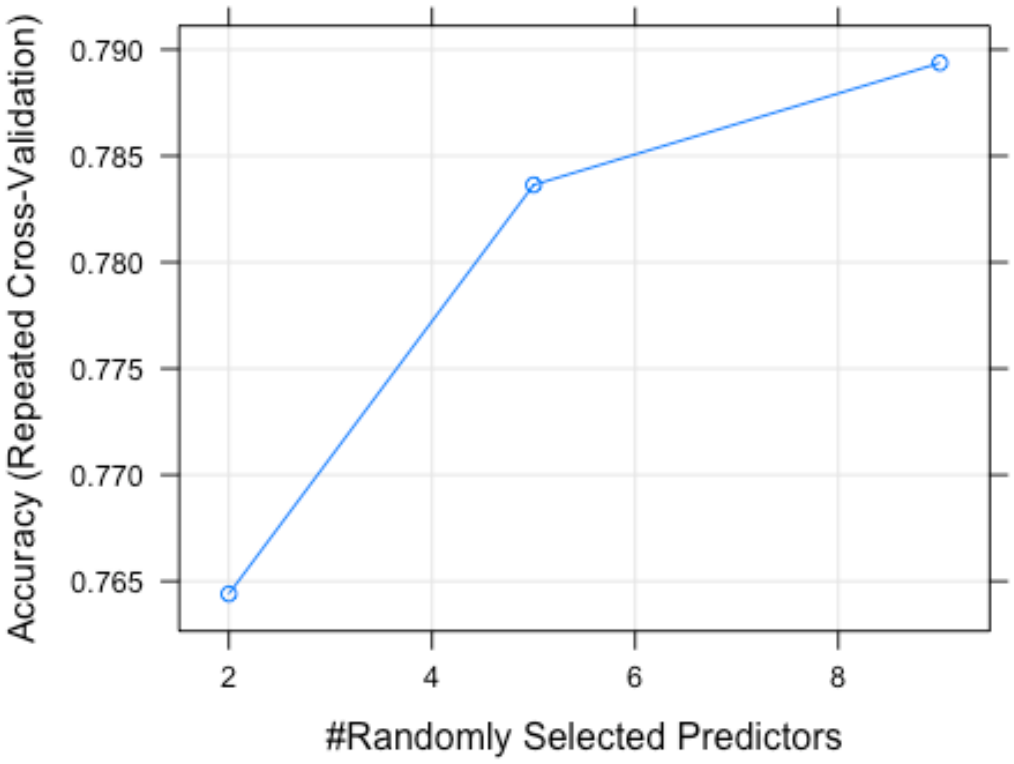


Figure 33 - Random Forest Result Plot of Accuracy and Number of Selected Predictors

The results of the random forest are better than those of the decision tree! We can see that we can reach almost 79% of accuracy in predictions when we use 8 predictors for the random forest. Now, let us see the most important features from the Random Forest.

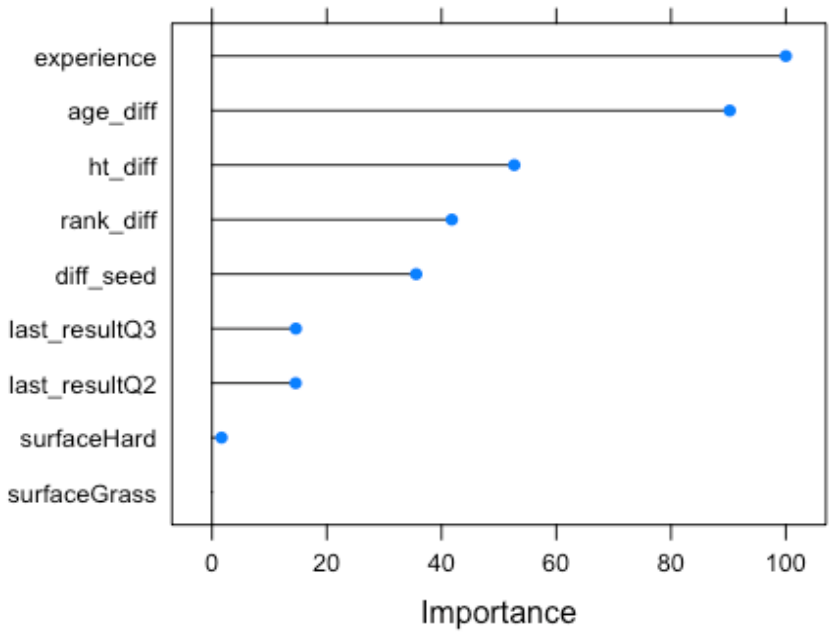


Figure 34 - Feature Importance Plot from Random Forest Model

The results of the feature importance tell that experience is the most important feature followed by the age difference of the players.

An LDA model was done next and its results are similar to the accuracy of the above models: accuracy of 74.62%. At 95% confidence interval, the lower and upper bounds are 66.24% and 81.84%, respectively.

Its confusion matrix results are seen below.

Prediction	-1	1
-1	0	0
1	33	97

Table 9 - Confusion Matrix Results of LDA Model

Then, K- Neighbor Classifier is done and the results are seen below.

K fold	Accuracy
5	0.7184126
7	0.7318756
9	0.7414169

Table 10 - KNN Model Accuracy for Different Number of K-folds

Not very different from the results of the previous models. At 9 K folds, it gives a decent accuracy of 74.14%.

Finally, a different model is applied to see if we can luckily get better results. Extreme Gradient Boosting model is applied; it is a technique that produces prediction models in form of decision trees. The results of this model are seen below, starting with the confusion matrix.

Prediction	-1	1
-1	9	0
1	24	97

Table 11 - Confusion Matrix Results of Extreme Gradient Boosting Model

This model gave the best results so far: 81.54% accuracy and lower and upper bounds of 73.79% and 87.8%, respectively, on the 95% confidence intervals. The next plot shows the feature importance.

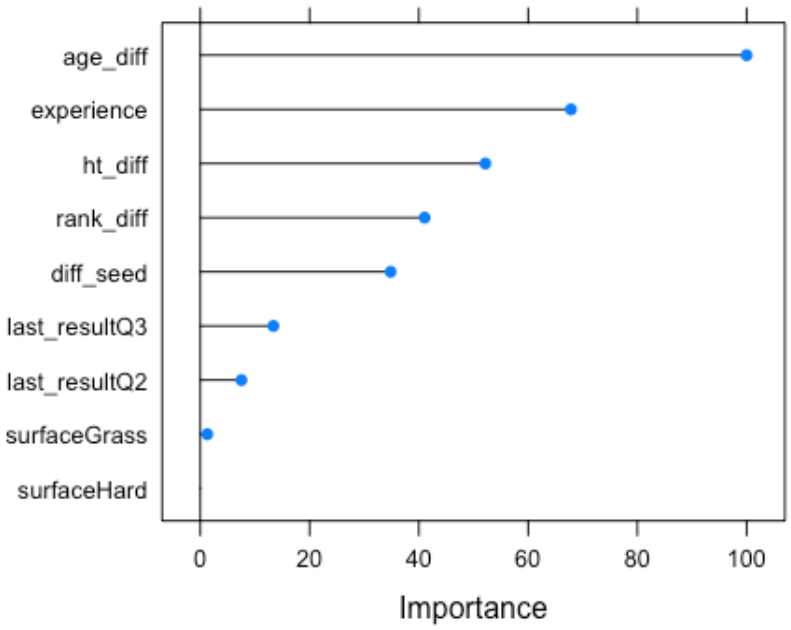


Figure 35 - Feature Importance from Extreme Gradient Boost Model

This model shows that the age difference is the most important feature used in the model.

CHAPTER 4. DISCUSSIONS

In the beginning of the work, it was explained that the main aim was to understand the performances of WTA players and predict the results of their matches. To this end, the objectives were set to accept or reject some hypotheses and the methods and results were explained in their respective sections. In the bigger scope, it is important to understand that the work aimed at using the available data and additionally some scraped data from official tennis websites but this is not all the data that can affect the performances and results. From the analysis of the work, not only was the descriptive understanding of the work done to answer the suggested questions, but also some models were used and eventually tested to see their accuracy in predictions of the results; thereby, fulfilling the aims of the work.

To discuss the resulting work for the first objectives, a summary of it is added. The main objective was to understand how anthropometric data can affect the performances and results of tennis. The attributes addressed here were the height and ages of the players; experience was derived from their age and years of play. For this, it was seen that taller players do, in fact, have an advantage over shorter players, in the game of tennis as they seemed to perform better. This was seen in the correlation of the height

differences with better career rankings where it was discovered that overall, taller players reach the lower – better – rankings, more, in comparison to shorter players. For taller players, the advantage was also in the number of aces they performed and the general winning percentage for taller players always being higher. Previous studies did not correlate the number of aces with winning for women; it was only correlated for men [8]. The analysis of the study did not go into comparing that, but it was curious to understand if taller players can perform better, in terms of aces, than shorter players and so it can only be inferred those aces affect wins since the players' winning percentages were higher.

As is obvious, the game of tennis is becoming much more attractive over the years. This is in-line with better performances, trainings, understandings, and the evolution of the game. In fact, one important factor may be that players' average heights are increasing over the years as was seen in this work. This was supported also by the analysis of checking for the effect of every 1 cm increase in height with winning percentage and the correlation proved height a relevant factor.

For age, most of the work only supported common hypotheses of certain age groups like 25-30 to be correct. The effect of age did not prove an important factor in the last sets of the games or in the finals of

tournaments; rejecting the hypothesis of older players performing better due to calmer nerves. In fact, it makes more sense that younger players would do equally well because they compensate for the nerves with more activity. The work also showed how experience is not really linked with age and that the optimal years of experience are 4 years regardless of which age the player becomes professional. This debunks the hypotheses of age having any specific advantage or disadvantage.

In the next objective, the aim to study different tournaments by looking at their different surfaces was interesting. The Roland Garros that is played in clay surface has significantly less aces happening there than the US Open, the Australian Open, and Wimbledon which have hard and grass surfaces. All the analysis regarding serves points and early winning points, including aces, happened most in Wimbledon with grass surface, US Open and Australian Open next, and the least in the Roland Garros. This suggested that players take longer rallies on the clay surface because the sand slows down the ball allowing for better receiving. However, the players must also be affected by the surface since they are running more distances more difficultly. This idea is addressed by Kovacs in one of his mentioned works [6], and would be an idea for future work.

From the context of the third objective where the focus was on the psychology of players and how it can change performances and results, a lot of work can be done using different data. From the data from tennis matches, regardless of how much match data can be added, the psychology of players can only be looked at in certain performances and matching it with why this happened. These performances were chosen during the occurrences of comebacks and upsets. Starting with upsets, these did not change when splitting them on different surfaces. This explains that an upset is a phenomenon that occurs due to different factors, probably mental resilience. Upsets, as explained, happen when the higher ranked player beats the lower ranked player. It was proved that at lower rank differences, upsets happen more. Since it is more difficult to win at bigger rank differences, it can be concluded that better players win because they are better.

However, it can also be said that better players do not lose their calm and do not underestimate their opponents because they mostly lose to other players only when their rankings are close. This effect happens even less in Grand Slams, as the analysis compared upsets in bigger versus smaller tournaments and it was seen that this happens even less. More evidence on this is that when upsets happen, it takes a player 3 sets – rather

than 2 sets – on most occasions, to beat a better player! The other phenomenon that was observed to support this was comebacks where a player loses the first set and comes back to win the match by winning the next two. This didn't prove to be a very common phenomenon, surprisingly; it happens in only 15% of matches. Comebacks also correlate with players that have less wins. This tells us that in women tennis, players are composed and a player that requires a comeback, is usually a player with less wins. From tennis data, occurrences of mental fatigue are not that high; this could also be because the matches have 3 sets only, not 5. Not enough is studied about mental fatigue and how it affects players of tennis. This may be, as mentioned, due to limited data that is specific to this. In this study by Dosil, interesting ways to address mental fatigue are tackled as well as the relevance and importance of calm mental states [9]. However, it is not common to find studies that objectively explain the phenomenon.

The models did pretty well when predicting the results of the games of tennis and the summary table below compares the accuracy for each.

Model	Accuracy
Logistic Regression	74.62%.
Linear Disc. Analysis Model	74.62%.
Random Forest Model	78.80%
K-neighbor Classifier Model	74.14%
Decision Trees Model	74.62%.
Extreme Gradient Boosting Model	81.54%.

Table 12 - Summary Table for the Applied Models and their Respective Accuracies

From the above table, the accuracy results of the logistic model, the LDA model, the random forest, the KNN, and the decision trees are seen. They were all around the 75% mark. The only model that exceeded that was the extreme gradient boosting model with an accuracy level of 85%. The age difference and the years of experience were the most important parameters for the Gradient boosting model, the same were for the Random Forest, and logistic regression found the last 3 results with the years of experience the most important.

CHAPTER 5. CONCLUSION

This work aimed to identify what affects performances in women tennis matches and to predict the results of these matches. It was seen that anthropometric data – mainly height – has an effect on tennis performances, and that on different surfaces, the athletes' type of play is different, giving advantage to a category of players per surface. Mental resilience of tennis players could be further researched and worked on pending some extra data. It was seen that better ranked players tend to do better in bigger tournaments and are less affected by the pressure, in comparison to players with worse ranks; also, better players need to do less comebacks in comparison but the phenomenon only happens in around 15% of the matches. The models predicted the outcomes of the matches with accuracies ranging between 74% and up to 82%. Even the model with the lowest accuracy gave a pretty decent accuracy of 74.12%; this was the K Neighbor Classifier. The Random Forest model did well with 8 predictors, obtaining an accuracy of 78.80%. The decision tree model was done for comparison but its results were obviously not as good as the Random Forest model, and its accuracy was 74.62%. The model with the highest accuracy of 81.54% was the Extreme Gradient Boosting model. It can be argued that other forms of data that are still not fully available can

improve the accuracy and give other insightful predictions. These not only include the data about psychology but other data can be useful if obtained like data about social backgrounds and data about training. One thing that can surely be concluded is that statistical analysis is an important tool that can be applied differently to various works, and the sports field is one where statistical analysis proves to be very beneficial.

CHAPTER 6. FUTURE RESEARCH

This section addresses improvements and changes that may have resulted in better results as well as future work in the same direction of this work. It is important to note that there was missing data for important attributes that might have affected the results. If the attributes were any less important, they would have been removed, however, the work aimed to prove and disprove the hypothesis with what was there in addition to the scraped tennis data.

It is important to note that not only tennis stats and information should be used in analysis of sports, performances, and results. Other data that was not available and difficult to be scraped about all the players in all the years could have influenced change. This data includes social status of players in their younger years: do privileged players have more access to trainings and material? Other data includes marital status and whether or not the players are pregnant or have had a child. This is because the levels of hormones fluctuate and can affect the performances of the players, and hence, their final match results. In this work by Womack, the authors discuss birth control and hormonal regulation for women [10]; this is because these changes affect several factors of the athletes' performance. In addition to this type of missing information, extra information about the

athlete's training can be considered like the years of training, type and method of training, and even coaches.

For performance, different data can be tracked using different technologies like video analysis technology [11] and positioning systems [12] . Most of the work, including this one, is done on tennis stats but a lot more – like injury analysis – can be understood from movements, steps, and even slips. For sport psychology that was analyzed in this work, better data can give more meaningful and relatable results. For example, the resting heartrate before a match, the type of food consumed, and even the environment around the player can affect the psychology before a big match.

Another point to be studied is whether or not the results change and improve if the study considered only top ranked players or excluded a certain category of players. This would have limited the data and reduced the observations greatly, so it was ignored in this work. Also, one of the advantages of using sports analysis is for profitable betting, however up to 80% of tennis betting happens in-play [13] – during the match – but our models do not work like that; they only make predictions before the game.

To conclude, the future of sports analysis will be a game changer for sure. With what is currently done, a lot of sports and older

methodologies have drastically improved. In sports other than tennis where transfer of players happen, data analysis is being used as well! It is very interesting to see the analysis for curiosity purposes only, as well, since tennis is a game of many fans. This work brought a better understanding to what happens in matches that cannot be noticed with the viewers' eyes.

CHAPTER 7. BIBLIOGRAPHY

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CHAPTER 8. SUPPLEMENTAL MATERIAL AND RESOURCES

The scripts used for this work are in a public github repository that can be accessed from the link below:

https://github.com/hatday/CodeHDWtaTFM22